On Some Connections Between Computation of Integrals of Analytic Functions, Root finding, Integer Factorization and NP-Completeness

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Abstract

We introduce a real integration-based method for localizing real zeros of analytic functions through point evaluation of an associated analytic primitive. A positive analytic kernel concentrates near the zeros, and differences of primitive values measure the accumulated mass on candidate intervals. Under quantitative bounds separating the mass near a genuine zero from the background on zero-free intervals, these differences provide a certified localization criterion with controlled evaluation error. The method applies independently of the arithmetic examples whenever the required quantitative bounds can be established. We apply this construction to integer factorization and an NP-complete quadratic Diophantine decision problem. In each application, the arithmetic input determines only the numerical parameter values of an explicitly defined analytic family with fixed formula structure and number of parameters. The encodings are holonomic; the primitives are not, but each is a component of a fixed-dimension polynomial initial-value problem with input-dependent coefficients and initial data. Uniform separation estimates and local mass bounds yield polynomial-time Turing reductions from the arithmetic tasks to approximate point evaluation of the corresponding primitives. Deciding quadratic feasibility uses only two nonadaptive evaluation queries. The binary parameter lengths and the sufficient number of accuracy bits are linear in the input length. Consequently, a uniform evaluator taking time polynomial in the argument length and the number of accuracy bits on these requests would yield polynomial-time factorization and, for the quadratic family, imply P = NP. The computational complexity of evaluating the resulting primitives remains open.

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1 Introduction

This paper presents the following constructions and results.

1. We introduce a new method for localizing real zeros of analytic functions, based on an analytic transformation of the original function and the evaluation of an associated primitive.

2. We study trigonometric encodings of integer factorization and an NP-complete quadratic Diophantine problem, and establish differential representations of their associated kernels and primitives. In particular, the primitives are DD-finite and differentially algebraic (D-algebraic), and are components of autonomous polynomial initial-value problems whose dimension and degree are independent of the arithmetic input. We also discuss the relationship of the encodings to previously proposed constructions.

3. We prove polynomial-time reductions from these arithmetic tasks to approximate evaluation of explicitly defined analytic families with a fixed formula structure and a fixed number of numerical parameters.

The proposed localization method is of independent interest and can be used separately from the arithmetic applications considered here. It is designed to detect zeros on prescribed intervals and progressively refine their locations, including zeros at which the function does not change sign. The general construction relies on analytic properties of the function and applies to a broad class of analytic equations whenever the necessary quantitative bounds are available. In this paper, we restrict attention to its theoretical application to establishing the stated polynomial-time reductions. Its practical use in numerical root-finding algorithms constitutes a separate direction for future research.

Let f be a real-valued analytic function, not identically zero, on a neighbourhood of a compact search interval. No sign condition is imposed on f: we set $\varphi = f^2$. If a nonnegative analytic encoding is already available, it may instead be used directly as φ. Fix either

$$g(s) = \operatorname{erfc}(s) \qquad \text{or} \qquad g(s) = \exp(-s^2),$$

and define

$$q_\alpha(x) = g(\alpha\varphi(x)), \qquad H_\alpha(x) = \int_{x_0}^{x} q_\alpha(\xi)\, d\xi, \qquad \alpha > 0.$$

On the real search interval, the kernel q_α is positive, equals one at every zero of φ, and tends uniformly to zero on compact sets containing no zeros as α increases. The differences

$$H_\alpha(v) - H_\alpha(u) = \int_u^v q_\alpha(x)\, dx$$

measure the positive integral mass on the intervals under consideration. Once the masses of admissible intervals containing a zero and those containing none are quantitatively separated, comparison of these differences with a suitable threshold permits successive localization by interval subdivision. This criterion also applies when the original function does not change sign at a zero.

The present construction differs from contour-integral approaches to locating zeros of analytic functions. Argument-principle methods, including the classical method of Delves and Lyness [1], use the logarithmic derivative f'/f, while other contour-based methods use the

reciprocal $1/f$; see, for example, Austin, Kravanja, and Trefethen [2]. In these formulations, zeros of f become poles of an auxiliary meromorphic function, and the integration contours must avoid these poles.

Methods using real integration also have established precedents. Ioakimidis [3] surveys quadrature methods, including constructions based on real-variable theory. Hungerbühler and Wasem [4] express the number of real zeros through an integral involving f, f', and f'', together with boundary terms, and develop variants that account for multiplicities. Their integrands can remain regular at real zeros, but preservation of the original complex domain of analyticity does not follow automatically. For example, their choice based on the Cauchy density gives

$$I_f(z) = \frac{f'(z)^2 - f(z)f''(z)}{\pi\big(f(z)^2 + f'(z)^2\big)}.$$

For the entire function $f(z) = z$, this becomes $1/(\pi(1+z^2))$, with poles at $z = \pm i$.

Certified subdivision algorithms based on approximate evaluation have also been developed for analytic functions. Yap, Sagraloff and Sharma [5] isolate clusters of complex roots of an analytic function by subdivision, using soft zero tests: predicates that are evaluated from approximations and whose positive answers are certified by quantitative margins. The integral-mass test of Section 2.4 shares the general philosophy of certified approximate predicates: the gap between the mass bounds B_0 and B_1 certifies decisions made from approximate primitive values. The tests of [5] are of Pellet type and use approximations of the function and its derivatives in complex boxes. The present test uses only values of a primitive at the endpoints of real intervals, and it also detects real zeros at which the function does not change sign.

Our construction instead applies an entire transformation to a nonnegative analytic encoding, without dividing by that encoding or using its derivatives in the kernel, so no new singularities arise (see below). For the arithmetic encodings studied below, this mechanism yields polynomial-time reductions to evaluation of explicit analytic families of fixed form.

An essential property of the method is that it preserves analyticity and introduces no new singularities within the domain of analyticity of the original function. Both choices of g are entire functions. Consequently, for every fixed α, the kernel q_α is holomorphic on any complex domain on which φ is holomorphic, and it admits a holomorphic primitive on every simply connected subdomain. In particular, if $\varphi(d) = 0$, then

$$q_\alpha(d) = H'_\alpha(d) = 1.$$

Thus, the zeros of the original function remain regular points of both the kernel and its local primitive. If φ is entire, both the kernel and its primitive are entire. For a function with pre-existing singularities, these statements apply within its domain of analyticity.

Small positive values of φ may also produce appreciable integral mass. Increasing α suppresses this background while narrowing the peaks near genuine zeros. Reliable localization therefore requires an upper bound on the mass of an admissible zero-free interval, a lower bound on the mass near a zero, and control of errors in the evaluated primitive values. These conditions determine the applicability of the general construction and the required computational resources. For the arithmetic families considered here, all necessary bounds are established explicitly.

The first arithmetic task is nontrivial divisor search: given an integer $t \geq 2$, find d satisfying

$$1 < d < t, \qquad d \mid t,$$

or determine that no such divisor exists. Repeated application of this procedure yields a complete prime factorization. The second task is to decide whether the equation

$$am^2 + bk = c, \qquad a, b, c \in \mathbb{Z}_{>0}, \qquad m, k \in \mathbb{Z}_{\geq 0},$$

has a solution. This decision problem is NP-complete by the theorem of Manders and Adleman [6].

The corresponding nonnegative analytic encodings are

$$f_t(x) = \sin^2\left(\frac{\pi t}{x}\right) + \sin^2(\pi x),$$

and

$$F_{a,b,c}(x) = \sin^2(\pi x) + \sin^2\left(\pi \frac{c - ax^2}{b}\right).$$

For $t \geq 4$, the zeros of f_t on $[2, t-1]$ are exactly the nontrivial divisors of t; smaller inputs are handled directly. The zeros of $F_{a,b,c}$ on $[0, \sqrt{c/a}]$ are precisely the feasible values of m, with

$$k = \frac{c - am^2}{b}.$$

Each sum of squares expresses two integrality conditions. The factorization encoding was obtained independently and is closely related to an earlier construction of Mateus and Vieira [7]; this relationship is discussed below. For these two encodings, one may use $\varphi = f_t$ or $\varphi = F_{a,b,c}$ directly in the general construction.

Fixing the choice of g gives two target families:

$$\mathcal{H}^{\mathrm{fac}}(t, \alpha; x) = \int_2^x g\left(\alpha \left[\sin^2\left(\frac{\pi t}{\xi}\right) + \sin^2(\pi \xi)\right]\right) d\xi,$$

and

$$\mathcal{H}^{\mathrm{quad}}(a, b, c, \alpha; x) = \int_0^x g\left(\alpha \left[\sin^2(\pi \xi) + \sin^2\left(\pi \frac{c - a\xi^2}{b}\right)\right]\right) d\xi.$$

Here x is the evaluation argument, while the numerical parameter lists are (t, α) and (a, b, c, α), respectively. The proved reductions use

$$\alpha = 4t^4 \quad \text{for factorization,} \qquad \alpha = 100D^2 \quad \text{for the quadratic problem,}$$

where

$$N = \left\lfloor \sqrt{c/a} \right\rfloor, \qquad D = b^2 + a^2(2N+1)^2.$$

Thus, the transformation parameter α is determined entirely by the original arithmetic data.

Each family is specified by a single fixed formula. As the arithmetic instance varies, the number of parameters, the number of trigonometric terms, and the composition structure

remain unchanged; only the numerical parameter values and their binary encoding lengths vary. In this sense, the dependence of the analytic construction on the original problem is quantitative. For fixed admissible parameters, the quadratic kernel and its primitive are entire functions of the argument. The factorization family is real analytic for $x > 0$. The singularity of the original factorization encoding at 0 lies outside the real intervals used in the construction; at the encoded divisors, both the kernel and the primitive are regular.

For each fixed instance, the primitive is moreover a component of the solution of an autonomous polynomial initial-value problem whose dimension, seven or eight, and polynomial degree do not depend on the arithmetic input; only its coefficients and initial data do (Proposition 3). This makes precise the sense in which the families have a fixed differential form. It also places their evaluation within the complexity theory of polynomial initial-value problems, where solution complexity has been studied in terms of trajectory length and related intrinsic quantities [8, 9]. For the explicit systems constructed here, the coefficient-weighted length-type quantity entering the general polynomial-IVP evaluation bound is $2^{\Theta(n)}$, where n is the binary length of the arithmetic input, on the evaluation requests generated by our reductions, although the integration interval and the actual arc length need not be exponential on every sequence of inputs (Section 5.9). By Theorem 6 and Corollary 1, a uniform evaluator whose total running time is polynomial in $\mathrm{size}(x) + p$ on these requests would yield polynomial-time integer factorization and, for the quadratic family, imply P = NP.

This differential formulation also admits a geometric reading of the whole construction. An arithmetic instance determines the real zero set of the encoding; the kernel $q_\alpha = g(\alpha\varphi)$ turns this set into a concentration profile; the primitive H_α accumulates its mass along the real axis; and, by Proposition 3, H_α is one coordinate of a trajectory of a polynomial vector field of fixed dimension and degree. Section 6 discusses how analytic continuation extends this picture. The complexity results rely only on the explicit quantitative estimates proved in the paper.

To construct the reductions, we use blocks of integer candidates

$$J_{L,R} = \left[L - \frac{1}{2}, R + \frac{1}{2}\right], \qquad L, R \in \mathbb{Z}, \qquad L \le R.$$

The half-integer endpoints leave a neighbourhood around every zero contained in the block. On zero-free blocks, a nonzero integer residual gives a uniform lower bound on the encoding function. Near a genuine zero, a quadratic upper bound on that function ensures sufficient width of the transformed peak. Theorem 5 combines these estimates into an explicit choice of scale, a separating integral-mass threshold, and sufficient accuracy for evaluating the primitive. This makes it possible to test for the presence of a zero using approximate endpoint values with a guarantee of correctness.

For a positive integer z, let

$$\ell(z) = 1 + \lfloor \log_2 z \rfloor$$

denote its binary length. For an arbitrary integer m, define

$$\lambda(m) = \ell\big(\max\{1, |m|\}\big) = 1 + \big\lfloor \log_2\big(\max\{1, |m|\}\big) \big\rfloor .$$

Thus $\lambda(m)$ counts the binary digits of $|m|$, with $\lambda(0) = 1$; the sign is encoded separately. For positive integers, $\lambda(m) = \ell(m)$.

A dyadic number is a rational number of the form

$$d = \frac{m}{2^k}, \qquad m \in \mathbb{Z}, \quad k \in \mathbb{Z}_{\geq 0}.$$

For dyadic input arguments, we use the reduced representation: m is odd whenever $k > 0$, and zero is represented by $m = k = 0$. The numerator and denominator are written explicitly in binary. We use the size measure

$$\mathrm{size}(d) = \lambda(m) + k + 1,$$

where $k + 1$ is the binary length of the denominator 2^k. Including signs and delimiters changes this size measure by at most a constant factor.

Let $n = \ell(t)$ for factorization and $n = \ell(a)+\ell(b)+\ell(c)$ for the quadratic problem. A request for p bits of absolute accuracy means obtaining an approximation $\widetilde{H}(x)$ satisfying

$$\left|\widetilde{H}(x) - H(x)\right| \leq 2^{-p}.$$

With the scales specified above,

$$\log_2 \alpha = O(n), \qquad p = O(n).$$

In particular, $p = 3n + 7$ suffices for factorization, and $p = 3n + 15$ suffices for the quadratic problem. The numerical value of the scale may be exponentially large, while the length of its binary representation remains linear.

The main complexity-theoretic result, stated in Theorem 6, establishes polynomial-time Turing reductions to approximate evaluation of the constructed analytic families. Deciding quadratic feasibility requires at most two nonadaptive queries. Finding one quadratic solution or one nontrivial divisor requires $O(n)$ queries, while complete prime factorization requires $O(n^2)$ queries. All constructed parameters, thresholds, query arguments, and oracle answers in the chosen binary format have $O(n)$-bit representations. All computation outside the oracle takes polynomial time. The reductions are correct for every oracle satisfying the required absolute error bound.

The sufficient accuracy and the number of queries needed to find one solution do not depend on the total number of zeros. The positive mass lower bound uses a neighbourhood of a single zero, and any mass contributed by other zeros is nonnegative. Finding one solution follows a single branch of the interval subdivision. For complete prime factorization, the procedure is applied recursively to the resulting cofactors. Enumerating all divisors or all solutions is a separate task whose output size may be superpolynomial.

In particular, evaluation of the quadratic family is NP-hard under polynomial-time Turing reductions, even at the specified scales and with only linearly many bits of absolute accuracy requested. The complexity of evaluating the primitives themselves remains open. By Corollary 1, for either family it suffices to have a single evaluation algorithm and one fixed polynomial P, independent of the arithmetic input, such that every request prescribed by Theorem 6 is answered in

$$P(\mathrm{size}(x) + p)$$

bit operations. The arithmetic parameters are exact integers. Their description lengths are $O(n)$ and are already controlled by the prescribed precision $p = 3n + 7$ or $p = 3n + 15$.

Here p specifies only the absolute error of the computed primitive value. Such an evaluator would give polynomial-time algorithms for the corresponding arithmetic tasks.

Integration is a classical source of hardness in the complexity theory of real functions. For a polynomial-time computable function on $[0,1]$, the indefinite integral can be #P-hard, and integration preserves polynomial-time computability for all such functions if and only if FP = #P (equivalently, P = $\mathrm{P}^{\#\mathrm{P}}$) [10]; see also [11]. For a fixed polynomial-time computable analytic function on a compact interval, by contrast, the primitive is again polynomial-time computable [12], [11, p. 208]. In these results, as in fast multiple-precision evaluation of holonomic functions [13], the function and its parameters are fixed and the cost is measured in terms of the requested precision; for holonomic functions, the constants depend on the differential equation. Evaluation of a fixed holonomic function can even be made uniform in the evaluation point, with relative precision away from its zeros [14]; the equation, and with it the constants, remain fixed. The present setting is different. The formula is fixed, the precision is linear in the input length, and hardness of the kind established here can arise only through the uniform dependence on the binary lengths of the parameters, as required in (6.1) and discussed in Section 6. Moreover, the primitives considered here are not holonomic (Proposition 4).

Connections between arithmetic problems and integration of analytic functions have classical precedents. Cosine product integration, problem AN14 in Garey and Johnson [15], takes an integer sequence $(a_1, \ldots, a_r)$ and asks whether the integral of the corresponding cosine product over a period is zero. After rescaling the interval, expansion of the cosines into complex exponentials gives

$$\int_0^1 \prod_{j=1}^{r} \cos(2\pi a_j x)\, dx = 2^{-r} \# \left\{ \sigma \in \{-1,1\}^r : \sum_{j=1}^{r} \sigma_j a_j = 0 \right\}.$$

Both the integrand and its primitive in AN14 are already entire functions. The displayed identity yields a gap of at least 2^{-r} between zero and positive integral values, so $O(r)$ bits of absolute accuracy suffice to distinguish the two cases. The product representation contains r cosine factors. Its expansion into 2^r exponential terms explains the counting identity but is not required to specify the function.

The distinguishing feature of our construction concerns its representation. In AN14, the length of the frequency list and the number of factors depend on the input. In our construction, both the trigonometric encodings and the associated primitive families have fixed formulas and fixed numbers of numerical parameters. By Proposition 3, the primitives are components of polynomial initial-value problems whose dimension and degree do not depend on the input, whereas the polynomial differential representation read off from the product form of the AN14 integrand has $2r$ trigonometric components. Arithmetic instances change only the parameter values and their binary encoding lengths. Arithmetic solutions correspond to real zeros of the encodings, and the localization method extracts information about these zeros from values of the corresponding primitives. The contribution lies in the combination of this localization procedure, a fixed analytic form, and explicit quantitative bounds establishing polynomial-time reductions.

For the factorization of odd semiprimes, Mateus and Vieira [7] used the function

$$h_t(z) = 1 - \cos\left(\frac{\pi t}{z}\right)\cos(\pi z).$$

They represented modular equations $r(z) \equiv 0 \pmod{s(z)}$ using the pair of functions

$$1 \pm \cos\left(\pi \frac{r(z)}{s(z)}\right) \cos(\pi z).$$

Their algorithms employ complex contour integration, residues, and the argument principle. In the construction involving $1/h_t$, zeros become poles, requiring control of the locations of singularities relative to the contours. The argument-principle approach also requires the determination of an appropriate branch of an analytic expression. The computational complexity of these procedures is left open.

In the method considered here, information about zeros is extracted from positive integral mass on real intervals. The transformation preserves regularity at the encoded solutions: the kernel equals 1 there, and the primitive remains analytic. Within each individual search, all queries concern a single primitive with a fixed base point and scale. Explicit mass-separation and evaluation-error bounds justify comparisons of its values and establish the stated polynomial-time reductions.

Section 2 develops the general localization method, its applicability conditions, and possible extensions. Sections 3–4 examine the arithmetic encodings. Section 5 proves the quantitative integral-mass separation bounds and polynomial-time reductions, and represents the primitives as components of polynomial initial-value problems of fixed dimension. Section 6 discusses the problem of evaluating the resulting analytic functions and possible geometric approaches to its solution.

2 Proposed method for root finding of analytic functions

This section develops a general method for localizing real zeros of analytic functions through values of an associated analytic primitive. The method is of independent interest and can be applied beyond the arithmetic encodings considered in this paper. Its integral-mass criterion applies to zeros of even multiplicity as well as to zeros at which the function changes sign.

A rigorous application requires quantitative separation between the transformed mass on zero-free intervals and the mass near genuine zeros, an admissible choice of interval boundaries and subdivision rules, and sufficient control of evaluation errors. In particular, the mass estimates must remain valid for zeros near interval endpoints, using one-sided neighbourhoods where appropriate. These requirements determine the scope of the method; the subsequent arithmetic applications provide explicit families for which they can be verified.

The entire transformations and polynomial modulations used below preserve analyticity within the original domain and introduce no singularities at the zeros being located. The resulting kernels and their local primitives remain regular there.

2.1 General outline and remarks

Let f be a real-valued analytic function on an open interval containing the compact interval $I = [X_L, X_R]$, and assume that f is not identically zero. We consider a method for locating

its real zeros by constructing an auxiliary analytic function

$$H_\alpha(x) = \int_{X_L}^{x} h\big(g\big(\alpha f(\xi)^2\big)\big)\, d\xi, \qquad \alpha > 0, \tag{2.1}$$

where $g = \operatorname{erfc}$. For the optional modulation, we consider the shifted power

$$h_{\beta,\rho}(s) = (s+\rho)^\beta, \qquad \rho \geq 0, \quad \beta \in \mathbb{Z}_{\geq 1}. \tag{2.2}$$

Here $h = h_{\beta,\rho}$, with its parameters suppressed in the notation for H_α. The choice $(\beta, \rho) = (1, 0)$ gives the identity modulation. When $\rho > 0$, the integral contains the known linear background $\rho^\beta(x - X_L)$; localization will use the mass above this background.

The construction consists of four steps:

1. **Initial transformation.** Set $q_\alpha(x) = g(\alpha f(x)^2)$. This function is analytic, takes values in $(0, 1]$ on I, and equals 1 at every zero of f. As α increases, its values tend uniformly to zero on compact subsets of I that contain no zeros of f.

2. **Modulation (optional).** Apply an analytic function h to q_α to modify the balance between the mass concentrated near the zeros and the remaining background. In the case of non-modulation we set $h(x) = x$.

3. **Integration.** Integrate the transformed function and subtract its known linear background. When the remaining background is sufficiently small relative to the mass near the zeros, the resulting function has a smooth step-like profile, with most of its growth concentrated near those zeros.

4. **Localization.** Use differences of the background-corrected values of H_α to test whether suitable subintervals contain zeros. Once this test is justified quantitatively, interval subdivision can be used for localization.

This section describes the general construction and identifies the estimates required for its rigorous use. The subsequent sections provide these estimates for the particular families considered in this article. Extensions to functions of several variables would require a corresponding localization procedure and separate quantitative analysis.

Recall the error function and the complementary error function,

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2)\, dt, \qquad \operatorname{erfc}(z) = 1 - \operatorname{erf}(z). \tag{2.3}$$

Both functions are entire. On the nonnegative real axis, erfc is positive and strictly decreasing, with $\operatorname{erfc}(0) = 1$ and $\operatorname{erfc}(y) \to 0$ as $y \to +\infty$ (Figure 1).

2.2 Step 1 – Initial transformation using the error function

For a parameter $\alpha > 0$, define

$$q_\alpha(x) = \operatorname{erfc}\big(\alpha f(x)^2\big)\,. \tag{2.4}$$

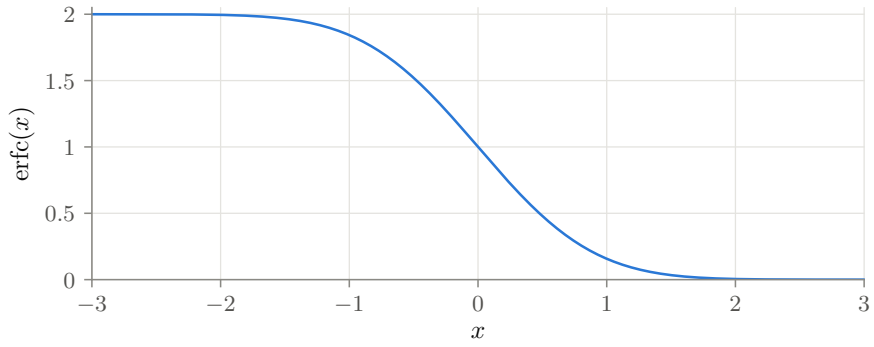


Figure 1: The complementary error function erfc(x) on the real axis. .

The square ensures that the argument of erfc is nonnegative on the real interval, even when f changes sign. Thus $0 < q_\alpha(x) \leq 1$, and $q_\alpha(d) = 1$ whenever $f(d) = 0$. For a compact set $K \subseteq I$ containing no zeros of f, the positive lower bound for $|f|$ on K implies uniform convergence $q_\alpha \to 0$ on K as $\alpha \to \infty$.

Remark 1. *The choice of the complementary error function is not essential to the method: one may also use the entire function* $g(t) = \exp(-t^2)$. *The quantitative estimates depend on the chosen transformation.*

If $f \geq 0$ throughout the interval, one may instead use

$$q_\alpha(x) = \operatorname{erfc}(\alpha f(x)) \,. \tag{2.5}$$

These two variants have different parameter scales and peak widths. The variant used will be specified in each application.

For illustration, consider

$$f(x) = \sin\left(\frac{\pi x}{2}\right) + \sin^2(4\pi x) + 0.1. \tag{2.6}$$

For this example, which takes both positive and negative values, we use the squared variant.

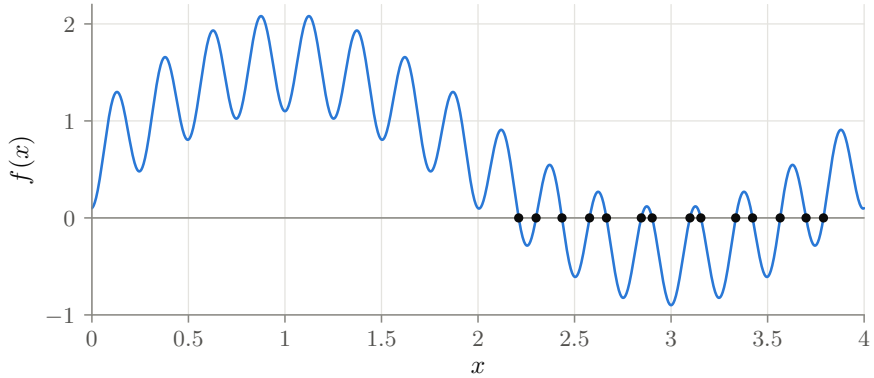


Figure 2: The function $f(x) = \sin(\pi x/2) + \sin^2(4\pi x) + 0.1$.

Increasing α suppresses the transformed values away from the zeros while preserving the value 1 at each zero. At the same time, the peaks become narrower and their integrated

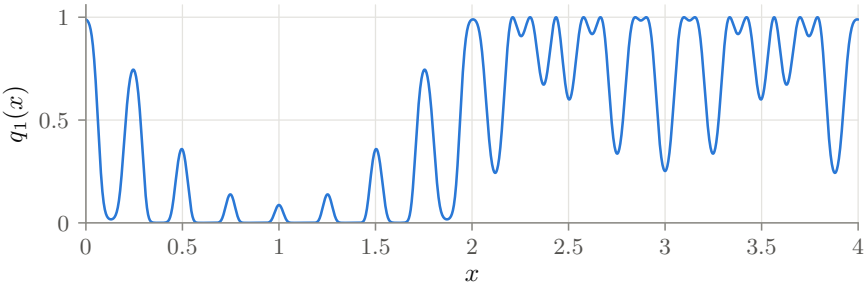


Figure 3: The transformed function $q_1(x) = \operatorname{erfc}(f(x)^2)$.

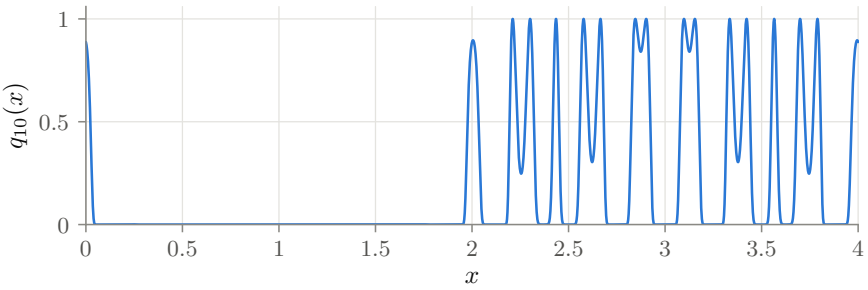


Figure 4: The transformed function $q_{10}(x) = \operatorname{erfc}(10f(x)^2)$.

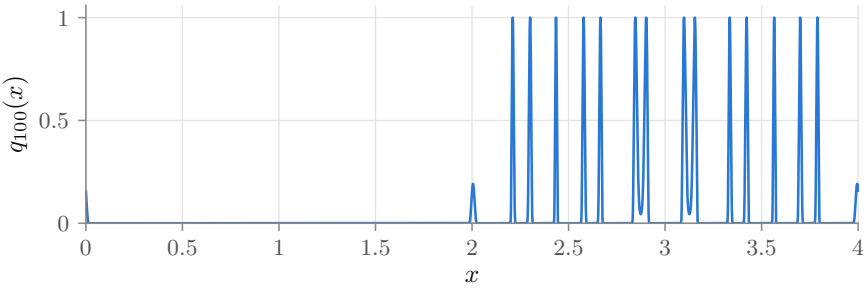


Figure 5: The transformed function $q_{100}(x) = \operatorname{erfc}(100f(x)^2)$.

masses decrease. Consequently, pointwise suppression alone is insufficient for reliable localization: the choice of α must also account for the mass remaining near a true zero and the accuracy of the subsequent integration.

Figures 2–5 above illustrate an example of applying this method. Figure 2 shows the test function $f(x) = \sin(\pi x/2) + \sin^2(4\pi x) + 0.1$ from (2.6), Figure 3 shows the result of the initial transformation, $q_1(x) = \operatorname{erfc}\big(f(x)^2\big)$, and Figures 4 and 5 show the effect of increasing the parameter to $\alpha = 10$ and $\alpha = 100$.

2.3 Step 2 (optional) – Modulation of the transformed function

The shifted-power modulation is defined by

$$p_\alpha(x) = h_{\beta,\rho}(q_\alpha(x)) = (q_\alpha(x) + \rho)^\beta, \qquad \rho \geq 0, \quad \beta \in \mathbb{Z}_{\geq 1}. \tag{2.7}$$

The integer exponent ensures analyticity, and no parity restriction is needed because $q_\alpha(x) + \rho > 0$ on the real interval. Its background value is ρ^β, while its value at a zero of f is $(1+\rho)^\beta$. It is therefore useful to distinguish p_α from its excess above the constant background:

$$\widehat{p}_\alpha(x) = (q_\alpha(x) + \rho)^\beta - \rho^\beta. \tag{2.8}$$

This excess is positive on I, and its value at every zero of f is $(1+\rho)^\beta - \rho^\beta$. For fixed ρ, β, it tends uniformly to zero on compact sets containing no zeros of f as $\alpha \to \infty$.

The effect of the modulation depends on both parameters. When $\rho = 0$ and $\beta > 1$, we have $\widehat{p}_\alpha = q_\alpha^\beta \leq q_\alpha$, so the integrated mass decreases. For $\rho > 0$, the shifted power can instead amplify the mass above the constant background. Indeed, for $\beta \geq 2$, the binomial expansion gives

$$\widehat{p}_\alpha(x) = \sum_{j=1}^{\beta} \binom{\beta}{j} \rho^{\beta-j} q_\alpha(x)^j \geq \beta \rho^{\beta-1} q_\alpha(x). \tag{2.9}$$

In particular, $\beta\rho^{\beta-1} \geq 1$ is a sufficient condition for the mass above the constant background not to decrease on any subinterval. This condition is not automatic for an arbitrary positive shift. The modulation also changes the residual background away from the zeros, so the useful peak mass and that residual background must be estimated together. For $\beta = 1$, subtracting the constant background gives $\widehat{p}_\alpha = q_\alpha$, irrespective of ρ.

2.4 Step 3 – Integration and interval subdivision

For the chosen transformation and modulation, define

$$H_\alpha(x) = \int_{X_L}^{x} p_\alpha(t)\, dt. \tag{2.10}$$

Its background-corrected counterpart is

$$\widehat{H}_\alpha(x) = H_\alpha(x) - \rho^\beta (x - X_L) = \int_{X_L}^{x} \widehat{p}_\alpha(t)\, dt. \tag{2.11}$$

For fixed finite parameters, both functions are real analytic and strictly increasing on I. The step-like description applies to $\widehat{H}_\alpha$, whose growth can be concentrated near the zeros. It does not imply discontinuities or exactly constant subintervals.

Let

$$Z = \{x \in I : f(x) = 0\}. \tag{2.12}$$

For an interval $J = [u, v] \subseteq I$, its transformed mass above the constant background is

$$\begin{aligned} M_\alpha(J) &= \widehat{H}_\alpha(v) - \widehat{H}_\alpha(u) \\ &= H_\alpha(v) - H_\alpha(u) - \rho^\beta(v - u) \\ &= \int_u^v \widehat{p}_\alpha(t)\, dt. \end{aligned} \tag{2.13}$$

The applications below will specify an admissible family of search intervals, used up to a prescribed stopping resolution, and parameters for which there are known bounds $0 \le B_0 < B_1$ such that

$$\begin{aligned} J \cap Z = \varnothing \quad &\Longrightarrow \quad M_\alpha(J) \le B_0, \\ J \cap Z \neq \varnothing \quad &\Longrightarrow \quad M_\alpha(J) \ge B_1. \end{aligned} \tag{2.14}$$

These bounds require control of the interval endpoints and of the mass in suitable neighbourhoods of the zeros, including one-sided neighbourhoods when appropriate.

Suppose that each required value of $\widehat{H}_\alpha$ is approximated with absolute error at most ε. If H_α is evaluated instead, the error and computational cost of the background correction must be included when obtaining these approximations. The corresponding approximation $\widetilde{M}_\alpha(J)$, obtained by subtracting two such values, then satisfies

$$\left|\widetilde{M}_\alpha(J) - M_\alpha(J)\right| \le 2\varepsilon. \tag{2.15}$$

If a threshold θ is chosen so that

$$B_0 + 2\varepsilon < \theta < B_1 - 2\varepsilon, \tag{2.16}$$

the test

$$\widetilde{M}_\alpha(J) > \theta \quad \Longleftrightarrow \quad J \cap Z \neq \varnothing \tag{2.17}$$

is correct for every admissible search interval. In particular, such a threshold exists whenever $B_1 - B_0 > 4\varepsilon$.

Starting from an admissible interval, first apply this test to determine whether it contains a zero. If it does, subdivide it according to a rule that preserves admissibility and covers its zeros by the child intervals, and retain a child interval for which the test is positive. The application must also guarantee a controlled decrease in interval size, as in binary subdivision. This procedure requires the presence of at least one zero, rather than uniqueness. It is continued until the stopping criterion of the application is met. If further numerical refinement is desired, a local method such as Newton's method may then be used under its convergence hypotheses, taking the multiplicity of the zero into account.

2.5 Quantitative requirements for the subsequent applications

To turn this construction into a rigorous procedure, the subsequent sections will establish:

1. a lower bound for the original function in absolute value on the admissible intervals that contain no zeros, sufficient to control the transformed background;
2. local estimates near a true zero, sufficient to give a positive lower bound for the transformed mass above the constant background;
3. choices of the transformation and modulation parameters, an absolute evaluation accuracy, including any background correction, and a decision threshold that separate the two cases;
4. an interval subdivision rule for which these estimates remain valid throughout the search.

For the complexity statements, the description lengths of all constructed parameters and query arguments, the requested number of bits of absolute precision, the number of oracle queries, and the computation performed outside the oracle must be bounded polynomially in the input length. These are requirements on the reduction. The complexity of evaluating the resulting family of analytic functions is a separate computational question.

2.6 Extension to zero-dimensional systems

The construction admits a natural extension to polynomial systems with finitely many solutions. For real polynomials $f_1, \dots, f_m$ in d variables, define

$$F(\mathbf{x}) = \sum_{j=1}^{m} f_j(\mathbf{x})^2, \qquad q_\alpha(\mathbf{x}) = \operatorname{erfc}\big(\alpha F(\mathbf{x})\big), \qquad \alpha > 0. \tag{2.18}$$

On $\mathbb{R}^d$, the zeros of F are precisely the common zeros of the system. The function q_α equals 1 at these points and converges uniformly to zero on compact sets containing no solutions.

Let $K \subset \mathbb{R}^d$ be a prescribed compact box in which solutions are to be located. For a subbox $B \subseteq K$, consider the background-corrected integral

$$M_\alpha(B) = \int_B \Big[\big(q_\alpha(\mathbf{x}) + \rho\big)^\beta - \rho^\beta \Big]\, d\mathbf{x}, \qquad \rho \geq 0, \quad \beta \in \mathbb{Z}_{\geq 1}. \tag{2.19}$$

The choice $(\beta, \rho) = (1, 0)$ corresponds to no modulation. Localization would proceed by subdividing K and using these integrals to detect subboxes containing solutions.

Complex solutions can be treated by writing $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ and

$$f_j(\mathbf{x} + i\mathbf{y}) = u_j(\mathbf{x}, \mathbf{y}) + i\, v_j(\mathbf{x}, \mathbf{y}), \tag{2.20}$$

where u_j and v_j are real polynomials. The real system $u_j = v_j = 0$, $j = 1, \ldots, m$, is equivalent to the original complex system. Thus, replacing F by

$$\widetilde{F}(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^{m} \left(u_j(\mathbf{x}, \mathbf{y})^2 + v_j(\mathbf{x}, \mathbf{y})^2 \right) \tag{2.21}$$

gives the corresponding construction on a compact box in $\mathbb{R}^{2d}$. The resulting kernel is real analytic in $(\mathbf{x}, \mathbf{y})$.

This extension requires separate quantitative analysis. Certified localization depends on a gap between the integral bounds for admissible boxes with and without solutions, together with control of solutions near subdivision boundaries. Zero-dimensionality alone does not provide these bounds. The required parameter sizes, evaluation precision, and number of queries must also be estimated before asserting a polynomial reduction.

3 Reformulation of factorization

Let $t \geq 2$ be an integer. Define the **factor function**

$$f_t(x) = \sin^2\left(\frac{\pi t}{x}\right) + \sin^2(\pi x), \qquad x \neq 0. \tag{3.1}$$

On the positive real axis, this nonnegative function encodes the divisors of t as its zeros.

Theorem 1 (Zeros and divisors)**.** *For every real $x \in [2, t]$,*

$$f_t(x) = 0 \quad \Longleftrightarrow \quad x \in \mathbb{Z}_{>0} \text{ and } x \mid t. \tag{3.2}$$

Proof. Both summands in (3.1) are nonnegative on the real axis, so their sum vanishes if and only if both summands vanish. Since $\sin(\pi r) = 0$ for real r if and only if $r \in \mathbb{Z}$, this is equivalent to

$$x \in \mathbb{Z}, \qquad \frac{t}{x} \in \mathbb{Z}. \tag{3.3}$$

Both integers are positive on the interval under consideration, which proves the assertion. □

The endpoint $x = t$ is a known zero corresponding to the trivial divisor t. For $t \geq 4$, searching on $[2, t-1]$ excludes this zero and gives precisely the nontrivial divisors. The cases $t = 2, 3$ can be handled directly.

A related trigonometric encoding was considered by Mateus and Vieira [7]. In their treatment of odd semiprimes, the relevant function is

$$1 - \cos\left(\frac{\pi t}{z}\right) \cos(\pi z). \tag{3.4}$$

The sum-of-squares formulation (3.1) enforces the two integrality conditions separately.

3.1 Analytic properties and holonomicity

The function f_t is analytic on $\mathbb{C} \setminus \{0\}$ and has an essential singularity at 0. It satisfies

$$f_t(\overline{z}) = \overline{f_t(z)}, \tag{3.5}$$

so its nonreal zeros occur in complex-conjugate pairs. The correspondence with divisors in Theorem 1 concerns real zeros.

Theorem 2 (Holonomicity)**.** *For each fixed $t \geq 2$, the function f_t is holonomic in x.*

Proof. The sine function is holonomic. Holonomic functions are closed under rational substitutions, products, and sums. Consequently,

$$\sin\left(\frac{\pi t}{x}\right), \qquad \sin(\pi x) \tag{3.6}$$

are holonomic, as are their squares and their sum. Here holonomicity is understood with respect to linear differential equations with polynomial coefficients over $\mathbb{C}$. □

Every zero of f_t in $[2, t]$ has multiplicity exactly two. Indeed, for a positive divisor $d \mid t$,

$$f_t(d) = f_t'(d) = 0, \qquad f_t''(d) = 2\pi^2 \left(1 + \frac{t^2}{d^4}\right) > 0. \tag{3.7}$$

Equivalently,

$$f_t(d+s) = \pi^2 \left(1 + \frac{t^2}{d^4}\right) s^2 + O(s^3) \qquad (s \to 0). \tag{3.8}$$

3.2 Alternative formulations

A formulation in terms of maxima is obtained by setting

$$f_{M,t}(x) = \cos^2\left(\frac{\pi t}{x}\right) + \cos^2(\pi x) = 2 - f_t(x). \tag{3.9}$$

Thus $0 \leq f_{M,t}(x) \leq 2$ for real $x \neq 0$, and on $[2, t]$ it attains the value 2 precisely at the divisors of t. Its holonomicity follows from the same identity.

An entire function of two variables encoding complementary factor pairs is

$$\mathcal{F}_t(x, y) = (t - xy)^2 + \sin^2(\pi x) + \sin^2(\pi y). \tag{3.10}$$

For $t \geq 4$ and real $(x, y) \in [2, t-1]^2$,

$$\mathcal{F}_t(x, y) = 0 \quad \Longleftrightarrow \quad x, y \in \mathbb{Z}_{>0}, \qquad xy = t. \tag{3.11}$$

Indeed, all three summands are nonnegative and must vanish simultaneously. The polynomial term imposes $xy = t$, while the trigonometric terms impose integrality. The function is entire on $\mathbb{C}^2$; the stated characterization applies to its real zeros in the specified square.

3.3 Complexity of function evaluation

Let $x = u/v \neq 0$ be an exactly specified rational number, where $u \in \mathbb{Z} \setminus \{0\}$ and $v \in \mathbb{Z}_{>0}$ are given in binary. Assume that the binary lengths of t, $|u|$, and v are at most n, where $n \geq 2$. We seek a dyadic approximation $\widetilde{f}$ satisfying

$$|\widetilde{f} - f_t(x)| \leq 2^{-n}. \tag{3.12}$$

Let $M_{\mathrm{mul}}(n)$ denote the bit complexity of multiplying two n-bit integers, with the usual regularity assumptions for fast integer arithmetic; in particular, $M_{\mathrm{mul}}(Cn) = O(M_{\mathrm{mul}}(n))$ for each fixed $C > 0$.

1. **Exact argument reduction.** Since $\sin^2(\pi r)$ has period 1 in r, compute

$$r_1 = \frac{tv}{u} - \left\lfloor \frac{tv}{u} \right\rfloor, \qquad r_2 = \frac{u}{v} - \left\lfloor \frac{u}{v} \right\rfloor. \tag{3.13}$$

   These are exact rational numbers in $[0, 1)$, and

$$f_t(x) = \sin^2(\pi r_1) + \sin^2(\pi r_2). \tag{3.14}$$

   The integer products and quotient–remainder operations involve $O(n)$-bit integers and require $O(M_{\mathrm{mul}}(n))$ bit operations using fast integer arithmetic. Thus, argument reduction is performed before any approximation of π is introduced.

2. **Evaluation of the elementary functions.** Compute dyadic approximations s_1, s_2, each with $O(n)$ bits, such that

$$|s_i - \sin(\pi r_i)| \leq \varepsilon, \qquad \varepsilon = 2^{-n-4}, \qquad i = 1, 2. \tag{3.15}$$

   For an explicit error budget, set $\delta = 2^{-n-8}$. Compute dyadic approximations $\widetilde{\pi}$, $\widetilde{r}_i$, y_i, and s_i satisfying

$$|\widetilde{\pi} - \pi| \leq \delta, \qquad |\widetilde{r}_i - r_i| \leq \delta,$$
$$|y_i - \widetilde{\pi}\widetilde{r}_i| \leq \delta, \qquad |s_i - \sin y_i| \leq \delta, \qquad i = 1, 2.$$

   Since $0 \leq r_i < 1$ and $|\widetilde{\pi}| < 4$,

$$\begin{aligned} |y_i - \pi r_i| &\leq |y_i - \widetilde{\pi}\widetilde{r}_i| + |\widetilde{\pi}|\,|\widetilde{r}_i - r_i| + |r_i|\,|\widetilde{\pi} - \pi| \\ &\leq 6\delta. \end{aligned}$$

   The sine function is 1-Lipschitz on the real line, hence

$$|s_i - \sin(\pi r_i)| \leq |s_i - \sin y_i| + |y_i - \pi r_i| \leq 7\delta < 2^{-n-4} = \varepsilon.$$

   These bounds specify the absolute errors of the computed intermediate values. The internal working precision of the elementary-function algorithms may be higher. Fast algorithms for π and the trigonometric functions give a total cost $O(M_{\mathrm{mul}}(n) \log n)$ [16, Sections 8 and 12].

3. **Post-processing and error control.** Compute the dyadic number

$$q = s_1^2 + s_2^2 \tag{3.16}$$

exactly. Since $|\sin(\pi r_i)| \le 1$,

$$|q - f_t(x)| \le 4\varepsilon + 2\varepsilon^2 < 2^{-n-1}. \tag{3.17}$$

Rounding q to the nearest multiple of 2^{-n} produces $\widetilde{f}$ with

$$|\widetilde{f} - f_t(x)| \le |\widetilde{f} - q| + |q - f_t(x)| < 2^{-n}. \tag{3.18}$$

The squarings, addition, and rounding require $O(M_{\mathrm{mul}}(n))$ bit operations.

Consequently, the factor function can be evaluated at exact rational inputs to absolute error at most 2^{-n} in

$$O(M_{\mathrm{mul}}(n) \log n) \tag{3.19}$$

bit operations, uniformly for inputs satisfying the stated binary-length bounds. The same bound holds for $f_{M,t} = 2 - f_t$. The uniformity in t follows from the explicit argument reduction and error analysis above.

3.4 Application of root finding

Theorem 1 converts the search for a nontrivial divisor into the search for a real zero of f_t on a suitable interval. Because f_t is nonnegative, ordinary bisection based on a change of sign cannot detect these zeros. Moreover, (3.7) shows that standard Newton iteration has only linear local convergence at them. The modified iteration

$$x_{k+1} = x_k - 2\frac{f_t(x_k)}{f_t'(x_k)} \tag{3.20}$$

has quadratic local convergence under the usual local hypotheses, but still requires a sufficiently accurate initial approximation.

Complex integration provides another possible formulation. If Γ is a positively oriented simple closed contour whose closure and interior avoid 0, and f_t has no zeros on Γ, the argument principle gives

$$\frac{1}{2\pi i}\int_\Gamma \frac{f_t'(z)}{f_t(z)}\,dz = N_\Gamma, \tag{3.21}$$

where N_Γ counts all enclosed zeros with multiplicity. Using this count to locate divisors requires control of the nonreal zeros as well. Residue-based formulations instead use meromorphic expressions such as $1/f_t(z)$. Under the same analyticity assumptions, the integral of $f_t(z)$ itself around Γ is zero.

Mateus and Vieira [7] reduce the factorization of semiprimes to particular contour-integration problems using residues and an extended form of the argument principle. They leave the time complexity of the proposed algorithms open.

The real-integral construction described in this article provides an alternative route: transform the nonnegative factor function into a kernel concentrated near its zeros, integrate this kernel, and use certified integral-mass tests to localize a divisor. Once a root is approximated with absolute error strictly less than $1/2$, rounding recovers the corresponding integer, which can be verified by exact division. Correctness and complexity of the integral-mass tests require the quantitative estimates developed in the subsequent sections.

4 Reformulation of an NP-complete problem

The same encoding principle applies to a classical NP-complete problem. Given positive integers a, b, c, represented in binary, consider the decision problem

$$\exists\, m, k \in \mathbb{Z}_{\geq 0} : \qquad am^2 + bk = c. \tag{4.1}$$

This problem is NP-complete by the theorem of Manders and Adleman [6]. Allowing zero coefficients adds only cases decidable by divisibility and integer-square-root tests, so restricting the coefficients to be positive preserves NP-completeness.

Set

$$T = \sqrt{\frac{c}{a}}, \qquad I_{a,c} = [0, T], \tag{4.2}$$

and define

$$F_{a,b,c}(x) = \sin^2(\pi x) + \sin^2\left(\pi \frac{c - ax^2}{b}\right). \tag{4.3}$$

For fixed a, b, c, this is an entire function of x, nonnegative on the real axis. It is also holonomic, by closure under polynomial substitutions, products, and sums.

Theorem 3 (Equivalence with quadratic Diophantine feasibility)**.** *For every real $x \in I_{a,c}$,*

$$F_{a,b,c}(x) = 0 \quad \Longleftrightarrow \quad x = m, \qquad \frac{c - am^2}{b} = k \quad \textit{for some } m, k \in \mathbb{Z}_{\geq 0}. \tag{4.4}$$

In particular, $F_{a,b,c}$ has a zero in $I_{a,c}$ if and only if (4.1) *has a solution.*

Proof. The two summands in (4.3) are nonnegative for real x. Their sum vanishes if and only if

$$x \in \mathbb{Z}, \qquad \frac{c - ax^2}{b} \in \mathbb{Z}. \tag{4.5}$$

The restriction $0 \leq x \leq \sqrt{c/a}$ makes both quantities nonnegative. Setting $m = x$ and $k = (c - am^2)/b$ gives $am^2 + bk = c$. Conversely, every nonnegative integer solution satisfies $m \in I_{a,c}$ and makes both sine terms vanish. □

4.1 NP-completeness of zero existence

The function family is represented by the binary triple (a, b, c) and the fixed formula (4.3). The same triple specifies the interval $I_{a,c}$ exactly. For the zero-existence decision problem, one may equivalently use $[0, N]$, where

$$N = \left\lfloor \sqrt{\frac{c}{a}} \right\rfloor = \left\lfloor \sqrt{\lfloor c/a \rfloor} \right\rfloor. \tag{4.6}$$

This integer endpoint can be computed in polynomial time, and the replacement preserves all zeros because they are integers.

Theorem 4 (NP-completeness for the encoded analytic family)**.** *Given the binary representation of $(a, b, c) \in \mathbb{Z}^3_{>0}$, deciding whether $F_{a,b,c}$ has a real zero in $I_{a,c}$ is NP-complete under polynomial-time many-one reductions.*

Proof. Constructing the fixed analytic expression (4.3) from (a, b, c) requires polynomial time and produces a description of length $O(L)$, where L is the total input length. Theorem 3 therefore gives a polynomial-time reduction from (4.1), proving NP-hardness.

For membership in NP, a certificate is a nonnegative integer m satisfying

$$am^2 \le c, \qquad b \mid (c - am^2). \tag{4.7}$$

Such an m has $O(L)$ bits because $m \le \sqrt{c/a}$. Both conditions can be checked by exact integer arithmetic in time polynomial in L. When they hold, $k = (c - am^2)/b$ is a nonnegative integer, and the preceding theorem supplies the required zero. □

This theorem concerns zero existence for the explicitly encoded family (4.3). Its short certificates are the integer solutions of the underlying Diophantine problem.

Remark 2. *Since both summands in* (4.3) *are nonnegative on the real axis, the real zeros of $F_{a,b,c}$ are exactly the integers m with $am^2 \equiv c \pmod b$. This set is symmetric under $m \mapsto -m$ and invariant under $m \mapsto m + b$, because $a(m + b)^2 - am^2 = b(2am + ab)$. Its elements $m \in \{0, \dots, N\}$ are precisely the feasible values in* (4.1)*, while elements $m > N$ correspond to $k < 0$. Hence* (4.1) *has a solution if and only if the least nonnegative real zero of $F_{a,b,c}$ exists and is at most N, and computing this least zero, or correctly reporting that no nonnegative real zero exists, is at least as hard as deciding* (4.1) *[6]. This periodic description also yields a constant-coefficient recurrence for the ordered global zero sequence. Let*

$$\mathcal{R} = \{r \in \{0, \dots, b-1\} : ar^2 \equiv c \pmod b\}.$$

If $\mathcal{R}$ is nonempty, write its elements as $r_0 < \cdots < r_{s-1}$ and let $(z_j)_{j \ge 0}$ be the increasing sequence of distinct nonnegative real zeros of $F_{a,b,c}$. Then

$$z_{ks+j} = kb + r_j \qquad (k \ge 0,\ 0 \le j < s),$$

so

$$z_{j+s} = z_j + b, \qquad z_{j+s+1} - z_{j+s} - z_{j+1} + z_j = 0 \qquad (j \ge 0).$$

Thus the global zero sequence is C-finite, with a recurrence of order at most $s+1$. Equivalently, if $R(u) = \sum_{j=0}^{s-1} r_j u^j$, then

$$\sum_{j\geq 0} z_j u^j = \frac{(1-u)R(u) + bu^s}{(1-u)(1-u^s)}.$$

The bounded feasibility problem asks whether this global sequence exists and its first term is at most N. This description is structural rather than algorithmic: the period s may be exponential in n, and the initial terms $r_0, \ldots, r_{s-1}$ are themselves solutions of the congruence.

4.2 Complexity of evaluating the encoding

Let $x = u/v$ be an exactly specified rational number, with $u \in \mathbb{Z}$ and $v \in \mathbb{Z}_{>0}$. Assume that the binary lengths of $a, b, c, |u|, v$ are at most n, where $n \geq 2$. Define the exact rational numbers

$$R_1 = \frac{u}{v}, \qquad R_2 = \frac{cv^2 - au^2}{bv^2}, \tag{4.8}$$

and reduce them modulo 1:

$$r_i = R_i - \lfloor R_i \rfloor, \qquad i = 1, 2. \tag{4.9}$$

Then

$$F_{a,b,c}(x) = \sin^2(\pi r_1) + \sin^2(\pi r_2). \tag{4.10}$$

All intermediate integer numerators and denominators have $O(n)$ bits. Exact reduction therefore costs $O(M_{\mathrm{mul}}(n))$ bit operations. The elementary-function evaluation and error analysis from Subsection 3.3 apply without change, giving a dyadic approximation $\widetilde{F}$ with

$$|\widetilde{F} - F_{a,b,c}(x)| \leq 2^{-n} \tag{4.11}$$

in $O(M_{\mathrm{mul}}(n) \log n)$ bit operations.

Thus the analytic encoding can be evaluated efficiently at specified rational arguments, while existence of a zero on the specified interval is an NP-complete decision problem.

4.3 From zero existence to analytic function evaluation

The reduction from this zero-existence problem to evaluation of the resulting analytic function is treated in Section 5, where the integral construction, the required accuracy, and the conditions for a polynomial reduction are collected.

5 Reduction to evaluation of the resulting analytic functions

This section applies the general localization method of Section 2 to the arithmetic encodings developed in Sections 3 and 4. The central step is to verify its quantitative hypotheses

uniformly over all candidate intervals used in the search. Bounds on the encoding away from its zeros, together with local estimates near genuine zeros, yield a certified integral-mass test based on approximate primitive values.

Theorem 5 supplies explicit scales, separating thresholds, and sufficient evaluation accuracies. Theorem 6 then combines this test with admissible interval subdivision and bounds the binary lengths of the queries, their number, and the computation required to construct and process them. This establishes polynomial-time Turing reductions from the arithmetic tasks to analytic function evaluation. Throughout each localization search, the transformation parameters and integration base point remain fixed.

5.1 The function associated with the quadratic encoding

Using the nonnegativity of $F_{a,b,c}$, set

$$q_\alpha(x) = g(\alpha F_{a,b,c}(x)), \qquad \alpha > 0, \tag{5.1}$$

where $g(s) = \operatorname{erfc}(s)$ or $g(s) = \exp(-s^2)$. For shifted-power modulation, define the background-corrected integral

$$\widehat{H}_{\alpha,\beta,\rho}(x) = \int_0^x \left[(q_\alpha(\xi) + \rho)^\beta - \rho^\beta\right] d\xi, \qquad \rho \geq 0, \quad \beta \in \mathbb{Z}_{\geq 1}. \tag{5.2}$$

The dependence on a, b, c is suppressed in the notation. The symbol ρ denotes the modulation shift, to distinguish it from the coefficient c of the Diophantine equation. When no modulation is applied, $(\beta, \rho) = (1, 0)$, corresponding to $h(s) = s$.

For every fixed choice of the coefficients and parameters, $\widehat{H}_{\alpha,\beta,\rho}$ is entire as a function of x. Indeed, $F_{a,b,c}$, g, and the integer-power modulation are entire, so the integrand is entire and has an entire primitive.

5.2 The function associated with factorization

For $t \geq 4$, the same construction applied to f_t on $[2, t-1]$ gives

$$q_\alpha^{(t)}(x) = g(\alpha f_t(x)) \tag{5.3}$$

and

$$\widehat{H}_{\alpha,\beta,\rho}^{(t)}(x) = \int_2^x \left[(q_\alpha^{(t)}(\xi) + \rho)^\beta - \rho^\beta\right] d\xi. \tag{5.4}$$

The parameters and the two choices of g are as above. For fixed parameters, this function is real analytic on an open interval containing $[2, t-1]$.

The integral construction transforms the nonnegative factor function into a kernel concentrated near its zeros and uses integral-mass tests to localize a divisor. Once a root is approximated with absolute error strictly less than $1/2$, rounding recovers the corresponding integer, which can be verified by exact division.

5.3 Integral-mass tests and required precision

For either resulting function, an admissible search interval $J = [u, v]$ has background-corrected mass

$$M(J) = \widehat{H}(v) - \widehat{H}(u), \tag{5.5}$$

where $\widehat{H}$ denotes the appropriate function from (5.2) or (5.4). Let Z be the set of zeros of its underlying encoding in the search domain. The required quantitative separation has the form

$$\begin{aligned} J \cap Z = \varnothing \quad &\Longrightarrow \quad M(J) \le B_0, \\ J \cap Z \neq \varnothing \quad &\Longrightarrow \quad M(J) \ge B_1, \end{aligned} \qquad B_0 < B_1. \tag{5.6}$$

The bounds may depend on the input and on the admissible interval. Their use requires control of zeros near interval boundaries. The initial search domain must itself be admissible or be covered by polynomially many admissible intervals. For localization, the subdivision rule must preserve admissibility, retain a zero whenever the parent interval contains one, and reach the required resolution within the stated query bound.

If each of the two required function values is known to absolute error at most ε, the computed mass $\widetilde{M}(J)$ satisfies

$$|\widetilde{M}(J) - M(J)| \le 2\varepsilon. \tag{5.7}$$

A threshold θ satisfying

$$B_0 + 2\varepsilon < \theta < B_1 - 2\varepsilon \tag{5.8}$$

therefore gives a correct zero-existence test on every admissible interval. Such a threshold exists whenever $B_1 - B_0 > 4\varepsilon$.

Writing $\Delta = B_1 - B_0 > 0$, a sufficient choice is

$$\varepsilon \le \frac{\Delta}{8}, \qquad p = \max\left\{0, \left\lceil \log_2 \frac{8}{\Delta} \right\rceil\right\}, \tag{5.9}$$

where each endpoint value is computed to absolute error at most 2^{-p}, and the threshold is $(B_0 + B_1)/2$. To obtain an input-dependent precision bound, we must therefore establish a quantitative gap for all intervals used in the search. The following construction supplies such a gap explicitly.

5.4 Search intervals with half-integer endpoints

All real zeros relevant to the two reductions are integers. We therefore search over blocks of integer candidates, using intervals

$$J_{L,R} = \left[L - \frac{1}{2}, R + \frac{1}{2}\right], \qquad L, R \in \mathbb{Z}, \quad L \le R. \tag{5.10}$$

Every zero in such an interval is at distance at least $1/2$ from its endpoints.

For factorization, the candidate block is $2, \ldots, t-1$, and the initial interval is

$$I_t^* = \left[\frac{3}{2}, t - \frac{1}{2}\right], \qquad t \ge 4. \tag{5.11}$$

For the quadratic problem, put

$$N = \left\lfloor \sqrt{c/a} \right\rfloor = \left\lfloor \sqrt{\lfloor c/a \rfloor} \right\rfloor \tag{5.12}$$

and use

$$I^*_{a,c} = \left[-\frac{1}{2}, N + \frac{1}{2} \right]. \tag{5.13}$$

These intervals have precisely the same relevant real zeros as the original domains. In the quadratic case, a zero in $I^*_{a,c}$ must be an integer $m \in \{0, \ldots, N\}$; hence $(c - am^2)/b \geq 0$, so every such zero represents a feasible solution. The definitions of the integral functions permit evaluation at all these endpoints.

If $L < R$, let $j = \lfloor (L + R)/2 \rfloor$. Subdivision uses

$$J_{L,j} \quad \text{and} \quad J_{j+1,R}, \tag{5.14}$$

whose common endpoint is $j + 1/2$. After a positive mass test, retain exactly one child with a positive test. If both children test positive, either may be chosen; the other is not explored when the goal is to find one zero. The number of subdivision levels is at most the ceiling of the base-two logarithm of the number of integer candidates. At a positive singleton block $J_{m,m}$, the integer m is the required zero and can be checked by exact arithmetic.

5.5 A uniform sufficient precision bound

The quantitative estimates and polynomial-time reductions below are established for the unmodulated case $h(s) = s$, equivalently $(\beta, \rho) = (1, 0)$. The shifted-power modulation introduced in Section 2 remains an optional part of the general construction, but is not used in these reductions. Write φ for the underlying nonnegative encoding and

$$H_\alpha(x) = \int_{x_0}^{x} g(\alpha\varphi(\xi))\, d\xi, \tag{5.15}$$

where x_0 is the base point of the appropriate integral. Thus H_α denotes the corresponding function $\widehat{H}_{\alpha,1,0}$.

Theorem 5 (Sufficient accuracy for integral-mass tests)**.** *Suppose that the search interval I^* is a block of integer candidates with half-integer endpoints, that $\varphi \geq 0$ on I^*, and that all zeros of φ in I^* are integers. Let $S \geq 8$ be an integer satisfying the following conditions:*

1. *$|I^*| \leq S$;*
2. *on every zero-free candidate-block interval $J \subseteq I^*$,*

$$\varphi(x) \geq S^{-1} \qquad (x \in J); \tag{5.16}$$

3. *for every zero $d \in I^*$ and $|s| \leq 1/2$,*

$$\varphi(d + s) \leq Ss^2. \tag{5.17}$$

Define

$$\nu = \lceil \sqrt{S} \rceil, \qquad \alpha = S^2, \qquad w = \frac{1}{2S\nu}, \qquad p = \lceil \log_2(32S\nu) \rceil . \tag{5.18}$$

Let

$$J = [u, v] = J_{L,R} = \left[L - \tfrac{1}{2}, R + \tfrac{1}{2} \right] \subseteq I^*, \qquad L, R \in \mathbb{Z}, \quad L \le R, \tag{5.19}$$

be any candidate-block interval. For either $g(z) = \operatorname{erfc}(z)$ or $g(z) = \exp(-z^2)$, the mass

$$M_\alpha(J) = H_\alpha(v) - H_\alpha(u), \qquad J = [u, v], \tag{5.20}$$

satisfies

$$\begin{aligned} J \cap Z = \varnothing \quad &\Longrightarrow \quad M_\alpha(J) \le w/2, \\ J \cap Z \neq \varnothing \quad &\Longrightarrow \quad M_\alpha(J) \ge w, \end{aligned} \tag{5.21}$$

where $Z = \{x \in I^* : \varphi(x) = 0\}$. Approximating each endpoint value of H_α to absolute error at most 2^{-p} therefore gives a correct test with threshold $3w/4$.

Proof. For either transformation,

$$g(z) \ge \frac{1}{2} \quad (0 \le z \le 1/4), \qquad g(z) \le \exp(-z^2) \quad (z \ge 1). \tag{5.22}$$

For the Gaussian, the first inequality follows from $\exp(-z^2) \ge 1 - z^2$, and the second is an equality. For the complementary error function,

$$\operatorname{erfc}(z) = 1 - \frac{2}{\sqrt{\pi}} \int_0^z \exp(-s^2)\, ds \ge 1 - \frac{2z}{\sqrt{\pi}} \ge \frac{1}{2} \quad (0 \le z \le 1/4), \tag{5.23}$$

and, for $z \ge 1$,

$$\begin{aligned} \operatorname{erfc}(z) &= \frac{2}{\sqrt{\pi}} \int_z^\infty \exp(-s^2)\, ds \le \frac{2}{z\sqrt{\pi}} \int_z^\infty s \exp(-s^2)\, ds \\ &= \frac{\exp(-z^2)}{z\sqrt{\pi}} \le \exp(-z^2). \end{aligned} \tag{5.24}$$

If J contains no zero, then $\alpha\varphi(x) \ge S$, whence

$$M_\alpha(J) \le S \exp(-S^2). \tag{5.25}$$

The exponential series gives $\exp(S^2) \ge S^4/2$. Moreover, $\nu \le S$ and $S \ge 8$, so $8\nu \le S^2$. Consequently,

$$M_\alpha(J) \le \frac{2}{S^3} \le \frac{1}{4S\nu} = \frac{w}{2}. \tag{5.26}$$

If J contains a zero d, the interval $[d - w, d + w]$ is contained in J, because $w \le 1/2$ and d is an integer. For $|s| \le w$,

$$\alpha\varphi(d + s) \le S^3 w^2 = \frac{S}{4\nu^2} \le \frac{1}{4}. \tag{5.27}$$

The integrand is therefore at least $1/2$ throughout this neighbourhood, giving

$$M_\alpha(J) \geq \int_{d-w}^{d+w} \frac{1}{2}\, dx = w. \tag{5.28}$$

This lower bound uses only one zero and is independent of how many additional zeros lie in J.

Finally, (5.18) implies

$$2^{-p} \leq \frac{1}{32S\nu} = \frac{w}{16}. \tag{5.29}$$

The error in the difference of two endpoint values is at most $w/8$. Thus a zero-free interval has computed mass at most $5w/8$, while an interval containing a zero has computed mass at least $7w/8$. The threshold $3w/4 = 6w/8$ separates the cases. □

The theorem supplies an explicit instance of the general criterion in Section 2.4: here $B_0 = w/2$, $B_1 = w$, and the midpoint threshold is $3w/4$. The chosen endpoint accuracy satisfies $2^{-p} \leq w/16$, leaving a strict margin between the two computed-mass cases. Thus the analytic estimates provide a certified zero-existence test on every admissible candidate block.

The sufficient precision is $p = O(\log S)$. Reusing endpoint values does not accumulate evaluation errors with the subdivision depth, since each test is formed from two endpoint approximations satisfying the same absolute error bound. Subsections 5.6–5.7 verify the hypotheses with parameters S whose binary lengths are linear in the arithmetic input length; Theorem 6 uses the resulting tests to establish the polynomial-time reductions.

5.6 Explicit accuracy for factorization

The following two propositions verify the quantitative hypotheses of Theorem 5 for the arithmetic encodings. The main step is to obtain a uniform lower bound on every zero-free candidate block. For this purpose, the two trigonometric phases are rounded to integers, and the resulting nonzero integer residual is bounded in terms of the rounding errors.

For the present encodings, explicit arithmetic identities yield bounds in terms of the input parameters. Combined with quadratic upper bounds near genuine zeros, these estimates determine sufficient transformation scales and show that a linear number of absolute-accuracy bits suffices for every interval test. We begin with the factorization encoding.

Proposition 1 (Explicit parameters and accuracy for factorization)**.** *Let $t \geq 4$ be an integer, and consider*

$$f_t(x) = \sin^2\left(\frac{\pi t}{x}\right) + \sin^2(\pi x)$$

on the candidate interval

$$I_t^* = \left[\frac{3}{2}, t - \frac{1}{2}\right].$$

Then the hypotheses of Theorem 5 hold with

$$S_t = 2t^2.$$

Consequently, the transformation parameter

$$\alpha_t = 4t^4$$

and the endpoint precision

$$p_t = \left\lceil \log_2\Big(64t^2 \left\lceil \sqrt{2}\, t \right\rceil\Big) \right\rceil$$

give a correct integral-mass test on every candidate-block interval $J \subseteq I_t^*$.

More explicitly, put

$$\nu_t = \left\lceil \sqrt{2}\, t \right\rceil, \qquad w_t = \frac{1}{4t^2\nu_t}.$$

For $J = [u, v]$, approximating each endpoint value of H_{α_t} with absolute error at most 2^{-p_t} and comparing their difference with $3w_t/4$ correctly determines whether J contains a nontrivial divisor of t.

If t has at most n binary digits, then

$$p_t \le 3n + 7.$$

Thus absolute error at most $2^{-(3n+7)}$ per endpoint value is sufficient for every mass test in the search.

Proof. The function f_t is nonnegative, and its zeros in I_t^* are precisely the nontrivial divisors of t. Moreover,

$$|I_t^*| = t - 2 \le S_t.$$

Let $J \subseteq I_t^*$ be a zero-free candidate-block interval, and let $x \in J$. Choose a nearest integer m to x within the candidate block and a nearest integer k to t/x. Write

$$x = m + u, \qquad \frac{t}{x} = k + v, \qquad |u|, |v| \le \frac{1}{2}.$$

At a half-integer endpoint, choose the nearest integer belonging to the block.

Since J contains no divisor of t, the integer $t - mk$ is nonzero. The identity

$$t - mk = ku + xv$$

and the bounds

$$x < t, \qquad k \le \frac{2t}{3} + \frac{1}{2} \le t$$

give

$$1 \le (t - mk)^2 \le 2t^2(u^2 + v^2).$$

Using

$$|\sin(\pi z)| \ge 2|z| \qquad (|z| \le \tfrac{1}{2}),$$

we obtain

$$f_t(x) \ge 4(u^2 + v^2) \ge \frac{2}{t^2} \ge \frac{1}{S_t}.$$

Now let $d \in \{2, \dots, t-1\}$ be a divisor of t, and let $|s| \le \frac{1}{2}$. Then

$$\left| \frac{t}{d+s} - \frac{t}{d} \right| = \frac{t|s|}{d|d+s|} \le \frac{t}{3}|s|.$$

Periodicity and the inequality $|\sin z| \le |z|$ yield

$$f_t(d+s) \le \pi^2 \left(1 + \frac{t^2}{9}\right) s^2 \le 2t^2 s^2 = S_t s^2,$$

where the last inequality follows from $\pi^2 < 10$ and $t \ge 4$.

All hypotheses of Theorem 5 therefore hold with $S = S_t$, which gives the stated scale, precision, and threshold. Finally,

$$\left\lceil \sqrt{2}\, t \right\rceil \le 2t,$$

so

$$p_t \le \left\lceil \log_2(128 t^3) \right\rceil.$$

If t has at most n binary digits, then $t < 2^n$, and hence $p_t \le 3n + 7$. □

5.7 Explicit accuracy for the quadratic encoding

Proposition 2 (Explicit parameters and accuracy for the quadratic encoding)**.** *Let a, b, c be positive integers, and define*

$$F_{a,b,c}(x) = \sin^2(\pi x) + \sin^2\left(\pi \frac{c - ax^2}{b}\right),$$

$$N = \left\lfloor \sqrt{\frac{c}{a}} \right\rfloor, \qquad D = b^2 + a^2(2N+1)^2, \qquad I^*_{a,c} = \left[-\frac{1}{2}, N + \frac{1}{2} \right].$$

Then the hypotheses of Theorem 5 hold with

$$S_{a,b,c} = 10D.$$

Consequently, the transformation parameter

$$\alpha_{a,b,c} = 100 D^2$$

and the endpoint precision

$$p_{a,b,c} = \left\lceil \log_2\left(320 D \left\lceil \sqrt{10D} \right\rceil\right) \right\rceil$$

give a correct integral-mass test on every candidate-block interval $J \subseteq I^*_{a,c}$.

More explicitly, put

$$\nu_{a,b,c} = \left\lceil \sqrt{10D} \right\rceil, \qquad w_{a,b,c} = \frac{1}{20 D \nu_{a,b,c}}.$$

For $J = [u, v]$, approximating each endpoint value of $H_{\alpha_{a,b,c}}$ with absolute error at most $2^{-p_{a,b,c}}$ and comparing their difference with $3w_{a,b,c}/4$ correctly determines whether there exist integers $m \in J \cap \{0, \ldots, N\}$ and $k \geq 0$ satisfying

$$am^2 + bk = c.$$

If each of a, b, c has at most n binary digits, then

$$p_{a,b,c} \leq 3n + 15.$$

Thus absolute error at most $2^{-(3n+15)}$ per endpoint value is sufficient for every mass test in the search.

Proof. The function $F_{a,b,c}$ is nonnegative. Its zeros in $I^*_{a,c}$ are integers $m \in \{0, \ldots, N\}$ for which

$$k = \frac{c - am^2}{b}$$

is a nonnegative integer. Furthermore,

$$|I^*_{a,c}| = N + 1 \leq D \leq S_{a,b,c}.$$

Let $J \subseteq I^*_{a,c}$ be a zero-free candidate-block interval, and let $x \in J$. Choose a nearest integer m to x within the candidate block and a nearest integer k to $y = (c - ax^2)/b$. Write

$$x = m + u, \qquad y = k + v, \qquad |u|, |v| \leq \frac{1}{2}.$$

Here k may have either sign.

The integer

$$\Delta = c - am^2 - bk$$

is nonzero. Indeed, if $\Delta = 0$, then

$$k = \frac{c - am^2}{b} \geq 0$$

because $0 \leq m \leq N$, giving a zero at $m \in J$, contrary to the choice of J.

Using

$$\Delta = a(x + m)u + bv$$

and $|x + m| \leq 2N + 1$, the Cauchy–Schwarz inequality gives

$$1 \leq \Delta^2 \leq D(u^2 + v^2).$$

Consequently,

$$F_{a,b,c}(x) \geq 4(u^2 + v^2) \geq \frac{4}{D} \geq \frac{1}{S_{a,b,c}}.$$

At a zero $m \in \{0, \dots, N\}$, write $k = (c - am^2)/b \in \mathbb{Z}_{\geq 0}$. For $|s| \leq \frac{1}{2}$,

$$\frac{c - a(m+s)^2}{b} - k = -\frac{a(2m+s)s}{b}.$$

It follows that

$$\begin{aligned} F_{a,b,c}(m+s) &\leq \pi^2 \left(1 + \frac{a^2(2N+1)^2}{b^2}\right) s^2 \\ &= \frac{\pi^2 D}{b^2} s^2 \leq 10 D s^2 = S_{a,b,c} s^2. \end{aligned}$$

Thus Theorem 5 applies with $S = S_{a,b,c}$, giving the stated scale, precision, and threshold.

If each of a, b, c has at most n binary digits, then

$$\begin{aligned} D &\leq b^2 + 4ac + 4a\sqrt{ac} + a^2 \\ &\leq b^2 + 6ac + 3a^2 < 10 \cdot 2^{2n}. \end{aligned}$$

Hence

$$S_{a,b,c} < 100 \cdot 2^{2n}, \qquad \left\lceil \sqrt{S_{a,b,c}} \right\rceil \leq 10 \cdot 2^n,$$

and therefore

$$32 S_{a,b,c} \left\lceil \sqrt{S_{a,b,c}} \right\rceil < 32000 \cdot 2^{3n} < 2^{3n+15}.$$

This proves $p_{a,b,c} \leq 3n + 15$.

The argument also covers $N = 0$, when the initial interval is $[-\frac{1}{2}, \frac{1}{2}]$ and the only integer candidate is 0. □

5.8 Polynomial-time reductions

For a positive integer z, write $\ell(z) = \lfloor \log_2 z \rfloor + 1$ for its binary length. Fix either $g(s) = \operatorname{erfc}(s)$ or $g(s) = \exp(-s^2)$. We use the unmodulated families with the scales from Subsections 5.6–5.7:

$$H_t^{\mathrm{fac}}(x) = \int_2^x g\big(4t^4 f_t(\xi)\big)\, d\xi, \qquad t \geq 4, \tag{5.30}$$

$$H_{a,b,c}^{\mathrm{quad}}(x) = \int_0^x g\big(100 D^2 F_{a,b,c}(\xi)\big)\, d\xi, \tag{5.31}$$

where f_t and $F_{a,b,c}$ are the encodings defined above, and

$$N = \left\lfloor \sqrt{c/a} \right\rfloor, \qquad D = b^2 + a^2(2N+1)^2.$$

An evaluation query specifies the arithmetic input, a dyadic argument x, and an integer precision $p \geq 0$. The oracle returns a signed binary integer q such that

$$\left| q\, 2^{-p} - H(x) \right| \leq 2^{-p}. \tag{5.32}$$

Thus approximations use a fixed-point representation with p fractional bits. The reductions below are correct for every choice of oracle answers satisfying (5.32). Each reduction is therefore a polynomial-time oracle algorithm for the corresponding evaluation problem. In particular, if O is an evaluation oracle for the quadratic family, then $\mathrm{NP} \subseteq \mathrm{P}^{O}$. All differences of dyadic endpoint approximations and all comparisons with rational thresholds are computed exactly using integer arithmetic.

Theorem 6 (Polynomial-time reductions to analytic function evaluation)**.** *For either fixed choice of g, the following statements hold.*

1. *Let $t \geq 4$ and $n = \ell(t)$. Finding a divisor d with $1 < d < t$, or correctly reporting that none exists, has a polynomial-time Turing reduction to evaluation of H_t^{fac}. The reduction uses at most*
$$2 + \lceil \log_2(t-2) \rceil = O(n)$$
oracle queries, each with $p = 3n+7$ bits of absolute accuracy. All query arguments are half-integers in $[3/2, t-1/2]$. A complete prime factorization can be obtained using $O(n^2)$ queries to the same family, with at most $3n+7$ accuracy bits per query. The inputs $t = 2, 3$ are handled directly.

2. *Let $a, b, c \in \mathbb{Z}_{>0}$ and $n = \ell(a) + \ell(b) + \ell(c)$. Deciding whether*
$$am^2 + bk = c, \qquad m, k \in \mathbb{Z}_{\geq 0},$$
has a solution requires at most two nonadaptive evaluation queries to $H_{a,b,c}^{\mathrm{quad}}$, each with $p = 3n+15$ bits of absolute accuracy, and polynomial-time processing of their answers. If a solution exists, one can find a pair (m, k) using at most
$$2 + \lceil \log_2(N+1) \rceil = O(n)$$
queries in total. All query arguments are half-integers in $[-1/2, N+1/2]$.

In both cases, all constructed parameters, thresholds, query arguments, and oracle answers have $O(n)$-bit representations, and all computation outside the oracle takes polynomial time in n. In particular, the scale descriptions satisfy $\ell(4t^4) \leq 4n+2$ and $\ell(100D^2) \leq 4n+14$ in the respective cases. Consequently, evaluation of the quadratic family is NP-hard under polynomial-time Turing reductions, even at the specified scales and with only linearly many bits of absolute accuracy requested.

Proof. Subsections 5.6 and 5.7 verify the hypotheses of Theorem 5 with $S = 2t^2$ and $S = 10D$, respectively. Put
$$\nu = \lceil \sqrt{S} \rceil, \qquad w = \frac{1}{2S\nu}.$$
The stated precision choices satisfy $2^{-p} \leq w/16$. For a candidate block
$$J_{L,R} = [L - 1/2, R + 1/2], \qquad L, R \in \mathbb{Z}, \quad L \leq R,$$
two endpoint approximations yield a computed mass $\widetilde{M}(J)$ with error at most $w/8$. By Theorem 5,
$$\widetilde{M}(J) \leq 5w/8 \quad \text{if } J \text{ contains no zero}, \qquad \widetilde{M}(J) \geq 7w/8 \quad \text{if } J \text{ contains a zero}.$$

Comparison with the rational threshold $3w/4$ therefore gives a correct zero-existence test.

Applying this test to the initial interval decides existence with two queries. If a zero exists, bisect the block of integer candidates and retain one child whose test is positive. Each subdivision requires only one new endpoint evaluation, since the two outer endpoint values are already known. A block with M candidates reaches a singleton after at most $\lceil \log_2 M \rceil$ subdivisions. Here $M = t-2$ or $M = N+1$, giving the claimed query counts. The singleton integer is a divisor in the first case; in the second case it gives m and $k = (c-am^2)/b$. These outputs are verified by exact integer arithmetic.

The quantities N, D, S, ν, the scales, and the rational thresholds can be constructed in polynomial time. Since $t < 2^n$ and, in the quadratic case, each coefficient is less than 2^n, the estimates $D < 10 \cdot 2^{2n}$ and $\alpha = 100D^2 < 2^{4n+14}$ give the asserted linear representation lengths. The factorization bound follows from $4t^4 < 2^{4n+2}$. Every query argument also has $O(n)$ bits. For the unmodulated primitives used in this theorem,

$$0 < g(\alpha\varphi(\xi)) \le 1$$

throughout the real integration intervals. Consequently, if x_0 is the fixed integration base point, then

$$|H(x)| = \left| \int_{x_0}^{x} g(\alpha\varphi(\xi))\, d\xi \right| \le |x - x_0|.$$

For factorization, the base point is 2, and every query argument belongs to $[3/2, t - 1/2]$. Since $t \ge 4$,

$$|H_t^{\mathrm{fac}}(x)| \le |x - 2| \le t - \frac{5}{2} < t.$$

For the quadratic family, the base point is 0, and every query argument belongs to $[-1/2, N+1/2]$. Hence

$$|H_{a,b,c}^{\mathrm{quad}}(x)| \le |x| \le N + \frac{1}{2}.$$

Thus we may take $B = t$ in the factorization case and $B = N + \frac{1}{2}$ in the quadratic case as bounds on the absolute values of the queried primitives. Since $t < 2^n$ in the first case and

$$N \le \sqrt{c/a} \le \sqrt{c}, \qquad c < 2^n,$$

in the second, both bounds satisfy

$$\log_2(B + 2) = O(n).$$

An oracle answer $q \in \mathbb{Z}$ satisfying

$$|q2^{-p} - H(x)| \le 2^{-p}$$

therefore obeys

$$|q| \le 2^p |H(x)| + 1 \le 2^p B + 1.$$

Its binary magnitude length is bounded by

$$\lambda(q) \le p + \lceil \log_2(B + 2) \rceil + 1 = O(n + p).$$

Since the reductions use $p = O(n)$, every oracle answer, including its sign, has $O(n)$ bits. Subtractions, threshold comparisons, and final verification therefore have polynomial bit cost.

For complete factorization, recursively split each composite cofactor using the divisor-search reduction. If there are r prime factors counted with multiplicity, then $2^r \le t$, so $r = O(n)$. The splitting tree has $O(n)$ nodes, each involving at most $O(n)$ queries on integers with at most n bits. Hence the total is $O(n^2)$. Finally, NP-hardness follows from the NP-completeness of the quadratic decision problem [6] and its two-query reduction established above. □

These conditions concern the reduction and its requested precision. The computational cost of evaluating the analytic functions themselves is a separate question.

Corollary 1 (Consequences of polynomial-time evaluation)**.** *Fix either choice of g. In each item it suffices that the stated guarantee and time bound hold on the evaluation requests generated by the corresponding reduction in Theorem 6.*

1. *If some algorithm, given an integer $t \ge 4$, a dyadic number x and an integer $p \ge 0$, returns an integer q satisfying* (5.32) *for $H = H_t^{\mathrm{fac}}$ in time polynomial in $\mathrm{size}(x) + p$, then a nontrivial divisor of t, or a correct report that none exists, and the complete prime factorization of t can be computed in time polynomial in $\ell(t)$.*
2. *If some algorithm, given $a, b, c \in \mathbb{Z}_{>0}$, a dyadic number x and an integer $p \ge 0$, returns an integer q satisfying* (5.32) *for $H = H_{a,b,c}^{\mathrm{quad}}$ in time polynomial in $\mathrm{size}(x) + p$, then* $\mathrm{P} = \mathrm{NP}$.

Proof. Let $n = \ell(t)$, respectively $n = \ell(a)+\ell(b)+\ell(c)$, as in Theorem 6. The scales $\alpha_t = 4t^4$ and $\alpha_{a,b,c} = 100D^2$ in (5.30) and (5.31) are determined exactly by the arithmetic input, can be computed from it in polynomial time, and are bounded above by fixed polynomials in the input values. Since $\log_2(4t^4) = 2 + 4\log_2 t$ and $D < 10 \cdot 2^{2n}$,

$$\ell(\alpha_t) \le 4n + 2, \qquad \ell(\alpha_{a,b,c}) \le 4n + 14.$$

In the reductions of Theorem 6, every query argument is a half-integer in $[3/2, t - 1/2]$, respectively in $[-1/2, N+1/2]$ with $N < 2^n$; hence $\mathrm{size}(x) \le n+3$. The requested precisions satisfy $p \le 3n + 7$, respectively $p \le 3n + 15$. Consequently, the complete input of every query has linear size:

$$\ell(t) + \ell(\alpha_t) + \mathrm{size}(x) + p \le n + (4n + 2) + (n + 3) + (3n + 7) = 9n + 12$$

for the factorization family, and

$$\begin{aligned}\ell(a) + \ell(b) + \ell(c) + \ell(\alpha_{a,b,c}) + \mathrm{size}(x) + p \\ \le n + (4n + 14) + (n + 3) + (3n + 15) = 9n + 32\end{aligned}$$

for the quadratic family. In particular, $\mathrm{size}(x) + p = O(n)$. By assumption, the algorithm satisfies the guarantee on each of these queries and answers it in time polynomial in n. The cofactors split off during complete factorization have at most n bits, so the same bounds apply to all their queries.

By Theorem 6, one nontrivial divisor is found with $O(n)$ queries, the complete prime factorization with $O(n^2)$ queries, and quadratic feasibility is decided with two queries; all remaining computation, including the exact integer comparisons with the thresholds, takes polynomial time. Replacing the oracle by the assumed algorithm therefore yields polynomial-time algorithms for these tasks. Finally, the quadratic feasibility problem is NP-complete [6]. If it can be decided in polynomial time, then P = NP. □

5.9 Polynomial differential representation

We record a representation of the kernels and primitives that makes their fixed form explicit. Here $\alpha > 0$ is arbitrary, the primitives are unmodulated, $H(x) = \int_{x_0}^{x} q(\xi)\,d\xi$ with $q = g(\alpha\varphi)$, and $x_0 = 0$ for the quadratic family and $x_0 = 2$ for the factorization family. The families (5.30) and (5.31) correspond to $\alpha = 4t^4$ and $\alpha = 100D^2$.

Proposition 3 (Polynomial differential representation)**.** *Fix the arithmetic parameters and $\alpha > 0$, and let $g(s) = \exp(-s^2)$ or $g(s) = \operatorname{erfc}(s)$.*

1. Quadratic family. *With $Y(x) = (c - ax^2)/b$ put*

$$u = \sin 2\pi X, \quad v = \cos 2\pi X, \quad s = \sin 2\pi Y(X), \quad r = \cos 2\pi Y(X),$$

$$\Phi = 1 - \tfrac{1}{2}(v + r), \qquad \Psi = \pi u - \tfrac{2\pi a}{b}\,Xs.$$

Then $\Phi = F_{a,b,c}$ and $\Psi = F'_{a,b,c}$. In the Gaussian case (X, u, v, s, r, q, H) solves

$$X' = 1, \quad u' = 2\pi v, \quad v' = -2\pi u, \quad s' = -\tfrac{4\pi a}{b} Xr, \quad r' = \tfrac{4\pi a}{b} Xs,$$
$$q' = -2\alpha^2 \Phi\Psi q, \quad H' = q.$$

In the complementary-error-function case the equation for q is replaced by

$$E' = -2\alpha^2 \Phi\Psi E, \qquad q' = -\tfrac{2\alpha}{\sqrt{\pi}}\,\Psi E,$$

where $E = \exp(-\alpha^2 F_{a,b,c}^2)$. The initial data at $x_0 = 0$ are $X = 0$, $u = 0$, $v = 1$, $s = \sin(2\pi c/b)$, $r = \cos(2\pi c/b)$, $E = \exp(-\alpha^2 \sin^4(\pi c/b))$, $q = g\big(\alpha \sin^2(\pi c/b)\big)$ and $H = 0$. The two systems have dimensions 7 and 8 and degree at most 4.

2. Factorization family. *With $u = \sin 2\pi x$, $v = \cos 2\pi x$, $\rho = 1/x$, $s = \sin(2\pi t\rho)$ and $r = \cos(2\pi t\rho)$,*

$$\rho' = -\rho^2, \quad s' = -2\pi t\rho^2 r, \quad r' = 2\pi t\rho^2 s,$$
$$\Phi = 1 - \tfrac{1}{2}(v + r) = f_t, \quad \Psi = \pi u - \pi t\rho^2 s = f_t',$$

together with $u' = 2\pi v$, $v' = -2\pi u$ and the same kernel and primitive equations, give autonomous polynomial systems in the variables (u, v, ρ, s, r, q, H) and $(u, v, \rho, s, r, E, q, H)$, of dimensions 7 and 8 and degree at most 5. The initial data at $x_0 = 2$ are $u = 0$, $v = 1$, $\rho = 1/2$, $s = 0$, $r = (-1)^t$, $E = \exp(-\alpha^2 f_t(2)^2)$, $q = g(\alpha f_t(2))$ and $H = 0$.

3. Consequently, in the Gaussian case

$$H'' + 2\alpha^2\varphi\varphi' H' = 0,$$

and in the complementary-error-function case

$$\varphi' H''' + \big(2\alpha^2\varphi(\varphi')^2 - \varphi''\big) H'' = 0.$$

The kernels and primitives are DD-finite and differentially algebraic.

The dimensions and degrees are independent of the arithmetic input; the coefficients and initial data carry the input dependence.

Proof. All equations follow by differentiation and the identity $\sin^2 y = \frac{1}{2}(1-\cos 2y)$; uniqueness for initial-value problems identifies the components with the specified functions. The system for $g = \operatorname{erfc}$ contains no division by φ' and is regular at $x_0 = 0$, where $F'_{a,b,c}(0) = 0$. The scalar equations follow from $q' = -2\alpha^2\varphi\varphi' q$ in the Gaussian case, by eliminating E from $q' = -(2\alpha/\sqrt{\pi})\varphi' E$ and $E' = -2\alpha^2\varphi\varphi' E$ in the other case, and from $H' = q$. The encodings are D-finite, so the coefficients of the scalar equations are D-finite, which gives DD-finiteness. Every component of the solution of a polynomial differential system is differentially algebraic, since the differential field it generates has finite transcendence degree; see also [17]. □

In the Gaussian case put $A = 2\alpha^2\varphi\varphi'$ and expand $A(x_0 + z) = \sum_{j\geq 0} A_j z^j$ and $H(x_0 + z) = \sum_{j\geq 0} h_j z^j$. Comparing coefficients in $H'' + AH' = 0$ gives

$$h_{j+2} = -\frac{1}{(j+1)(j+2)} \sum_{k=0}^{j} A_k (j-k+1) h_{j-k+1}, \qquad j \geq 0, \tag{5.33}$$

with $h_0 = H(x_0)$ and $h_1 = \exp(-\alpha^2\varphi(x_0)^2)$. At another expansion centre the same recursion uses the values of H and H' there. It is a convolution recurrence of growing length rather than a recurrence of fixed order. On the requests of Theorem 6 the factor α^2 in the coefficients A_j has bit length linear in n but numerical value exponential in n, so a certified evaluation method based on (5.33) must control truncation and rounding errors uniformly under this growth.

Remark 3. *The encodings $F_{a,b,c}$ and f_t are holonomic, and hence so is $A = 2\alpha^2\varphi\varphi'$. Consequently, the coefficients A_j in (5.33) satisfy a linear recurrence with polynomial coefficients whose order and degree can be bounded independently of the arithmetic input. The growing length of (5.33) comes from the convolution with the coefficients of H.*

Proposition 4 (Non-holonomicity of the kernels and primitives)**.** *Fix an admissible arithmetic input, a scale $\alpha > 0$ and either choice of g. Then the kernel $q_\alpha = g(\alpha\varphi)$ and the primitive H_α are not holonomic, for both encodings $\varphi = f_t$ and $\varphi = F_{a,b,c}$.*

Proof. A function h satisfying a linear differential equation with polynomial coefficients has a bound

$$|h(z)| \leq \exp\big(O(|z|^M)\big)$$

along any fixed vertical ray beyond the finitely many singular points, for some $M > 0$. This follows by writing the equation as a first-order system with rational coefficients and applying Gronwall's inequality. We show that the kernels violate this bound.

For the factorization encoding, as $y \to +\infty$,

$$f_t\Big(\frac{1}{4} + iy\Big) = \frac{i}{4} e^{2\pi y}\big(1 + o(1)\big), \qquad f_t\Big(\frac{1}{4} + iy\Big)^2 = -\frac{1}{16} e^{4\pi y}\big(1 + o(1)\big).$$

For the quadratic encoding, fix $\sigma > b/(2a)$ and, for all sufficiently large integers k, put

$$y_k = \sqrt{\sigma^2 - \frac{c}{a} + \frac{b}{a}\left(k + \frac{1}{4}\right)}, \qquad z_k = \sigma + iy_k, \qquad \kappa = \frac{4\pi a\sigma}{b} > 2\pi.$$

Then

$$\frac{c - az_k^2}{b} = k + \frac{1}{4} - i\frac{2a\sigma}{b} y_k,$$

and therefore

$$F_{a,b,c}(z_k) = -\frac{i}{4} e^{\kappa y_k}\big(1 + o(1)\big), \qquad F_{a,b,c}(z_k)^2 = -\frac{1}{16} e^{2\kappa y_k}\big(1 + o(1)\big).$$

Since $|\exp(-\alpha^2\varphi^2)| = \exp(-\alpha^2 \operatorname{Re}\varphi^2)$, it follows that

$$\Big|\exp\Big(-\alpha^2 f_t\big(\tfrac{1}{4} + iy\big)^2\Big)\Big| \geq \exp\Big(\frac{\alpha^2}{32} e^{4\pi y}\Big), \qquad \Big|\exp\big(-\alpha^2 F_{a,b,c}(z_k)^2\big)\Big| \geq \exp\Big(\frac{\alpha^2}{32} e^{2\kappa y_k}\Big)$$

for all sufficiently large y and k, respectively. Thus the Gaussian kernels grow doubly exponentially along the ray $z = \frac{1}{4} + iy$ in the factorization case and along the sequence (z_k) in the quadratic case.

The complementary-error-function kernels have the same growth. Indeed,

$$\operatorname{erfc}(w) \sim \frac{e^{-w^2}}{\sqrt{\pi}\, w} \qquad (|w| \to \infty)$$

uniformly on each closed subsector $|\arg w| \leq 3\pi/4 - \delta$, with $0 < \delta < 3\pi/4$; see [18, Eq. 7.12.1]. Along the ray and the sequence above, $\arg(\alpha\varphi)$ tends to $\pi/2$ or $-\pi/2$, so this asymptotic formula applies, and the factor $1/(\sqrt{\pi}\, w)$ decays only exponentially.

The resulting growth contradicts the bound for holonomic functions. Thus neither kernel is holonomic. Since differentiation preserves holonomicity and $H'_\alpha = q_\alpha$, neither primitive is holonomic either. □

Write one of the systems of Proposition 3 as

$$y' = Q(y), \qquad Q(Y) = \sum_\nu c_\nu Y^\nu, \qquad d = \deg Q, \qquad \Sigma_Q = \sum_\nu \|c_\nu\|_\infty.$$

For an evaluation point x, put $I_x = [\min\{x_0, x\}, \max\{x_0, x\}]$ and define

$$\operatorname{len}(x_0, x) = \int_{I_x} \|y'(\xi)\|_\infty \, d\xi, \qquad L(x_0, x) = \int_{I_x} \Sigma_Q \, \max\{1, \|y(\xi)\|_\infty\}^d \, d\xi.$$

The first quantity is the arc length; the second is the pseudo-length used in the general polynomial-IVP bound. The inequality

$$\|Q(Y)\|_\infty \le \Sigma_Q \max\{1, \|Y\|_\infty\}^d$$

gives $\mathrm{len}(x_0, x) \le L(x_0, x)$. When $x < x_0$, set $z(\tau) = y(x_0 - \tau)$; then $z' = -Q(z)$, with the same degree, coefficient norm, and lengths.

For rational coefficients and initial data, [8, Theorem 16] computes the solution with absolute error 2^{-p} in bit time polynomial in $\deg Q$, p, $L(x_0, x)$, $\log_2(2 + \|y(x_0)\|_\infty)$ and $\log_2(2 + \Sigma_Q)$, where the degree of this polynomial grows linearly with the dimension; here L is the pseudo-length of [8, Definition 12]. In [8] the coefficient norm of a polynomial vector field is the maximum over its components of the sums of the absolute values of their coefficients. It differs from Σ_Q by at most the factor $\dim y \le 8$, so the bounds on $L(x_0, x)$ below hold for either normalization. The systems of Proposition 3 have coefficients involving π; the extension of [8] to real coefficients and initial data given by oracles is [19, Theorem 7]. There the oracle represents the coefficients, initial values, and endpoints. For an ordinary bit algorithm, their required approximations must be generated internally from the exact arithmetic input, with that work included in the total evaluation cost. This introduces no additional input-accuracy parameters.

Lemma 5.1 (Pseudo-length on the reduction queries)**.** *Consider the systems of Proposition 3 at the scales $\alpha = 4t^4$ and $\alpha = 100D^2$ used in Theorem 6. For a query associated with an arithmetic instance t or (a, b, c), let n denote the binary length of that instance; during recursive factorization, this is the length of the current cofactor. Then, on every half-integer query of Theorem 6,*

$$\pi\alpha^2 \le L(x_0, x) \le 2^{O(n)}, \qquad \textit{hence} \qquad L(x_0, x) = 2^{\Theta(n)}.$$

Moreover, $\|y(x_0)\|_\infty = 1$ and $\log_2(2 + \Sigma_Q) = O(n)$.

Proof. For the upper bound, the interval length and Σ_Q are at most $2^{O(n)}$. The trigonometric components, q, and E when present are bounded in absolute value by one. In the quadratic system $X = \xi$; in the factorization system $\rho = 1/\xi$ is bounded on the integration intervals. Also, $|H(\xi)| \le |\xi - x_0|$. Thus

$$\sup_{\xi \in I_x} \|y(\xi)\|_\infty \le 2^{O(n)},$$

which gives $L(x_0, x) \le 2^{O(n)}$ because $d \le 5$. The initial values listed in Proposition 3 have maximum norm one.

For the lower bound, the equation for q in the Gaussian system, or for E in the complementary-error-function system, contains the monomial uq, or respectively uE, with coefficient $-2\pi\alpha^2$. Every query point is a half-integer and x_0 is an integer. Hence

$$\Sigma_Q \ge 2\pi\alpha^2, \qquad |x - x_0| \ge \frac{1}{2}, \qquad L(x_0, x) \ge \pi\alpha^2.$$

For factorization, $\alpha = 4t^4 = 2^{\Theta(n)}$. For the quadratic family, $2N + 1 \ge \sqrt{c/a}$ implies

$$D = b^2 + a^2(2N + 1)^2 \ge \max\{a^2, b^2, c\}.$$

Consequently $abc \le D^2$, and

$$n \le \log_2(abc) + 3 \le 2\log_2 D + 3, \qquad \alpha = 100D^2 \ge 100\,2^{n-3}.$$

This proves the lower bound. □

The actual arc length need not be exponential on every sequence of instances. For example, if $a = b = c = 2^k$, then $N = 1$ and

$$F_{a,b,c}(x) = \sin^2(\pi x) + \sin^2\bigl(\pi(1 - x^2)\bigr)$$

is independent of k. All integration intervals lie in $[-1/2, 3/2]$. This fixed nonconstant analytic function has finitely many monotonicity intervals there. On each of them, q, and E when present, have variation at most one. The remaining components have uniformly bounded variation as well, using $|H'| \le 1$. Thus $\mathrm{len}(x_0, x) = O(1)$ in this example, although $L(x_0, x) = 2^{\Theta(n)}$. On instances with $N \ge 2^{\Omega(n)}$, on the other hand, the arc length is exponential as well: the component $u = \sin 2\pi X$ has total variation 4 on every unit interval, so $\mathrm{len}(0, N + \frac{1}{2}) \ge 4N$. Similarly, $\mathrm{len}(2, t - \frac{1}{2}) \ge 4(t - 3)$ in the factorization case.

By Lemma 5.1, for the systems under consideration

$$\|y(x_0)\|_\infty = 1, \qquad \log_2(2 + \Sigma_Q) = O(n), \qquad \log_2(2 + L(x_0, x)) = O(n).$$

Therefore, a uniform evaluator whose total bit cost were polynomial in p, $\mathrm{size}(x)$, $\log_2(2 + \Sigma_Q)$, and $\log_2(2 + L(x_0, x))$ would answer the requests of Theorem 6 in polynomial time, since there $p = 3n + 7$ or $p = 3n + 15$. Corollary 1 would then give polynomial-time integer factorization for the factorization family and P = NP for the quadratic family. The estimate for L concerns this particular differential representation; it is not a lower bound for every algorithm evaluating the primitive.

More generally, polynomial ordinary differential equations are known to provide continuous characterizations of several classical complexity classes, including P, PSPACE and EXPTIME, under suitable geometric resource measures [9, 20, 21]. These results show that polynomial flows can support computational behaviour of widely different complexity when their available geometric resources are allowed to grow. They do not, however, determine the evaluation complexity of the particular systems of Proposition 3: those systems are not constructed to simulate a computation, but arise from a single explicit integral of an explicit entire, respectively real-analytic, function.

5.10 Number of divisors and output size

Let $\tau(t)$ denote the number of positive divisors of t, and let n be its binary input length. This number need not be polynomially bounded in n. More precisely, Wigert's maximal-order theorem (see, for example, [22]) implies

$$\max_{1 \le t < 2^n} \tau(t) = 2^{\Theta(n/\log n)}. \tag{5.34}$$

Thus the maximal number of divisors is superpolynomial but subexponential in the binary input length.

This growth does not alter the sufficient absolute accuracy for the integral-mass tests. The lower bound for a positive interval in Theorem 5 uses the neighbourhood of just one zero. Additional zeros can only increase the integral of the nonnegative kernel. The upper bound for a zero-free interval holds throughout that interval, including the tails of all nearby peaks, and does not require summing a separate error for each zero. Nor does a large zero count force a large representation of $H_\alpha(x)$: the integrand is at most one, so the magnitude of an integral value is bounded by the integration length. Consequently, the endpoint precision prescribed by (5.18) with $S = 2t^2$ remains sufficient without any factor depending on $\tau(t)$.

There are three different output tasks. Finding one nontrivial divisor requires only the single branch described above, hence $O(n)$ oracle evaluations. A complete prime factorization can be obtained by recursively applying this search to each composite factor. If there are k prime factors counted with multiplicity, then $2^k \le t$, so $k \le \lfloor \log_2 t \rfloor$. The resulting binary splitting tree has at most $2k-1$ nodes. All numbers processed have at most n bits, and the total number of oracle evaluations is therefore $O(n^2)$, with at most $3n+7$ absolute-accuracy bits requested per evaluation. The inputs 2 and 3 are handled directly; for larger inputs, a negative initial test certifies that no nontrivial divisor exists.

In contrast, explicitly enumerating all divisors requires writing $\tau(t)$ integers and hence at least $\Omega(\tau(t))$ output operations. Exploring every positive branch would address this larger task and does not have a polynomial bound in n alone. The reductions above establish bounds for one-divisor search and for the compact prime factorization, not for this enumeration.

Finally, a bound on the number and precision of oracle queries does not bound the cost of answering them. A numerical integration procedure that resolves every peak separately may have a cost depending on the number of divisors. No polynomial-time evaluation algorithm for H_α is established here.

6 Discussion

The first contribution of this work is a general method for localizing real zeros through values of analytic primitives. The method is of independent interest, applies without requiring a sign change at a zero, and preserves regularity at the zeros being located. Its rigorous use depends on quantitative mass separation, admissible interval boundaries, and controlled evaluation errors. Applications to other analytic encodings or to several variables require corresponding estimates.

The second contribution is the realization of this method by explicitly defined analytic families with fixed formula structures and fixed numbers of numerical parameters. Arithmetic instances select parameter values while preserving the number of trigonometric terms and the composition structure. The quadratic family is entire in its argument, and the factorization family is real analytic on the positive real axis. These concrete representations provide a basis for investigating evaluation through series expansions, analytic continuation, and the geometry of growth in the complex plane. Each primitive is moreover a component of the solution of an autonomous polynomial initial-value problem of dimension at most eight whose coefficients and initial data carry the arithmetic input (Proposition 3). These properties make the transformed primitives objects of differential algebra that can be studied

through their own equations. Proposition 3 supplies explicit differential representations, and (5.33) gives a constructive recurrence for local Taylor coefficients. This suggests evaluation algorithms that exploit the DD-finite representation, differential-algebraic identities, and analytic continuation. A concrete objective is to determine whether these representations permit sharper uniform bounds for coefficient generation, continuation, and error control. Corollary 1 specifies the evaluation bound under which such algorithms would also yield polynomial-time algorithms for the associated arithmetic tasks.

For the quadratic family, Remark 2 gives a C-finite description of the global nonnegative zero sequence of the encoding whenever this sequence is nonempty. For the unmodulated kernel and primitive of Section 5, these zeros also have a direct differential interpretation: for every real x and every $\alpha > 0$,

$$F_{a,b,c}(x) = 0 \quad \Longleftrightarrow \quad q_\alpha(x) = 1 \quad \Longleftrightarrow \quad H'_\alpha(x) = 1.$$

Thus, on the nonnegative real axis, this sequence consists exactly of the points at which the primitive has unit slope. The description in Remark 2 does not by itself provide a polynomial-time procedure for constructing the recurrence and its initial data; the explicit list of residues may have exponential size.

These representations allow the sequence of constructions to be viewed as a passage from a discrete arithmetic problem to a geometric one. At the first level, the arithmetic solutions are represented by the real zeros of an analytic encoding within the prescribed search interval. At the second level, the transformed kernel replaces this discrete set by a continuous concentration profile localized near the zeros, while integration converts this local concentration into an accumulated quantity accessible through values of the primitive. At the third level, the polynomial-IVP representation places this primitive within a trajectory of a finite-dimensional polynomial flow. The same arithmetic information therefore admits descriptions in terms of a zero set, a distribution of integral mass, and the geometry of a dynamical-system trajectory.

Complex continuation adds another level to this picture. In the complex domain, the relevant structure includes not only the locations of the real zeros, but also directions of growth and decay, the distribution of complex zeros and singularities, and the geometry of the paths along which the function is analytically continued and evaluated, together with the evolution of the corresponding trajectories of the polynomial flow. In the present work, this perspective serves as a structural interpretation and as motivation for further investigation. This makes it possible to formulate the evaluation problem not only in terms of the original integral or a local Taylor expansion, but also in terms of geometric properties of the corresponding continuations and flows. Determining which of these geometric quantities actually control the computational cost for the present families is a separate problem.

The third contribution consists of the polynomial-time Turing reductions from integer factorization and an NP-complete quadratic Diophantine problem to evaluation of these families. The mass-separation estimates make approximate primitive values sufficient for certified recovery of arithmetic information. Both the parameter description lengths and the sufficient number of absolute-accuracy bits are linear in the arithmetic input length.

To obtain a polynomial-time algorithm from these reductions, evaluation must be uniform in all input-dependent parameters. Write H_θ for the primitive, where θ includes the arithmetic input, the scale α, and any other varying parameters, including the integration base point

if it is not fixed. Let size$(\cdot)$ denote binary encoding length; for a rational evaluation point x, this includes the numerator and denominator. By Corollary 1, it suffices to have a single algorithm that, given (θ, x, p), returns a dyadic number $\widetilde{H}$ satisfying

$$\left|\widetilde{H} - H_\theta(x)\right| \le 2^{-p} \qquad \text{in time} \qquad P(\text{size}(x) + p)\,, \tag{6.1}$$

where P is one fixed polynomial, independent of the input values. A separate dependence on size(θ) is not needed: on the evaluation requests generated in Section 5 one has size$(\theta) = O(n)$ and $p = 3n + 7$ or $p = 3n + 15$, so the parameter descriptions are controlled by the requested precision. For the reductions, it suffices that this guarantee hold on all these requests. The parameter p concerns only the absolute error of the computed primitive value; it does not prescribe additional input accuracies for the exact arithmetic parameters.

Here polynomial dependence on accuracy means polynomial dependence on the number p of requested bits, equivalently on $\log_2(1/\varepsilon)$ for absolute error ε, rather than on $1/\varepsilon$. Likewise, the argument is measured by its binary length, and the numerical values of the arithmetic input and of the scale α must not enter the bound. For a single polynomial-time computable analytic function on a compact interval, the primitive is again polynomial-time computable [12], [11, p. 208]. This result is non-uniform in the sense of [23]: the algorithm, the degree of the polynomial, its constants and the accuracy from which the bound applies may depend on the function. Uniform complexity in this sense, with the argument and the parameters varying and measured by their bit lengths, was also studied by Du and Yap [24] for absolute approximation of hypergeometric functions, a class that contains erf. In contrast, (6.1) requires one algorithm and one polynomial P for all requests generated in Section 5 simultaneously, and the bound must hold at the precision $p = 3n+7$ or $p = 3n+15$ prescribed by the request, not only from some input-dependent accuracy onwards. A bound polynomial in p for each fixed parameter choice is insufficient if its constants or degree depend uncontrollably on those parameters. In particular, a cost polynomial in the numerical value of α or the interval length may be exponential in the arithmetic input length.

Possible computational approaches include certified quadrature, Taylor continuation with certified error control based on the recursion (5.33), numerical integration of the polynomial systems of Proposition 3, and alternative integral or series representations. Analyticity alone does not supply the required complexity estimate. The cost of resolving narrow peaks, constructing expansion coefficients, and controlling truncation and roundoff errors must be assessed uniformly over the parameter family. For the differential systems of Proposition 3, the pseudo-length in the general IVP evaluation bound is $2^{\Theta(n)}$ at the prescribed scales (Section 5.9), even though the actual arc length can remain bounded on some sequences of inputs. By Corollary 1, a uniform evaluator with total running time polynomial in size$(x) + p$ on the prescribed requests would yield polynomial-time integer factorization for the factorization family and P = NP for the quadratic family. The separation estimates proved here specify a sufficient output accuracy, but do not by themselves bound these computational costs. Identifying parameter regimes in which evaluation is polynomial and quantifying the cost outside them are further objectives.

We intend to investigate geometric approaches to evaluating the analytic primitives constructed here, drawing on the Gawronski–Müller–Reinhard (GMR) method and related work [25, 26]. In particular, we plan to explore modified GMR (MGMR) schemes combining locally chosen auxiliary functions with Taylor analytic continuation and variable step sizes. The central question is how to choose the auxiliary functions, continuation path, and step

sizes jointly, using complex-growth estimates to control truncation errors and cancellation. This choice must also account for the cost of constructing expansion coefficients and for error amplification when recovering the original primitive from the transformed representation.

The aim is to reduce the number of series terms and the working precision required for certified evaluation, thereby accelerating zero localization through faster computation of primitive values. Establishing the applicability of this approach to the present parameterized families, identifying parameter regimes in which it provides a computational advantage, and bounding the total cost remain tasks for future research. Any resulting total bit-complexity estimate must be uniform over the arithmetic inputs and include the costs of auxiliary-function construction and analytic continuation. Corollary 1 thus provides a precise criterion for such estimates: on the prescribed requests, where the exact parameter descriptions have size $O(p)$, a bound of the form (6.1) would yield polynomial-time algorithms for the arithmetic tasks.

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